The European XFEL – An X-ray laser driven by the world’s largest superconducting accelerator

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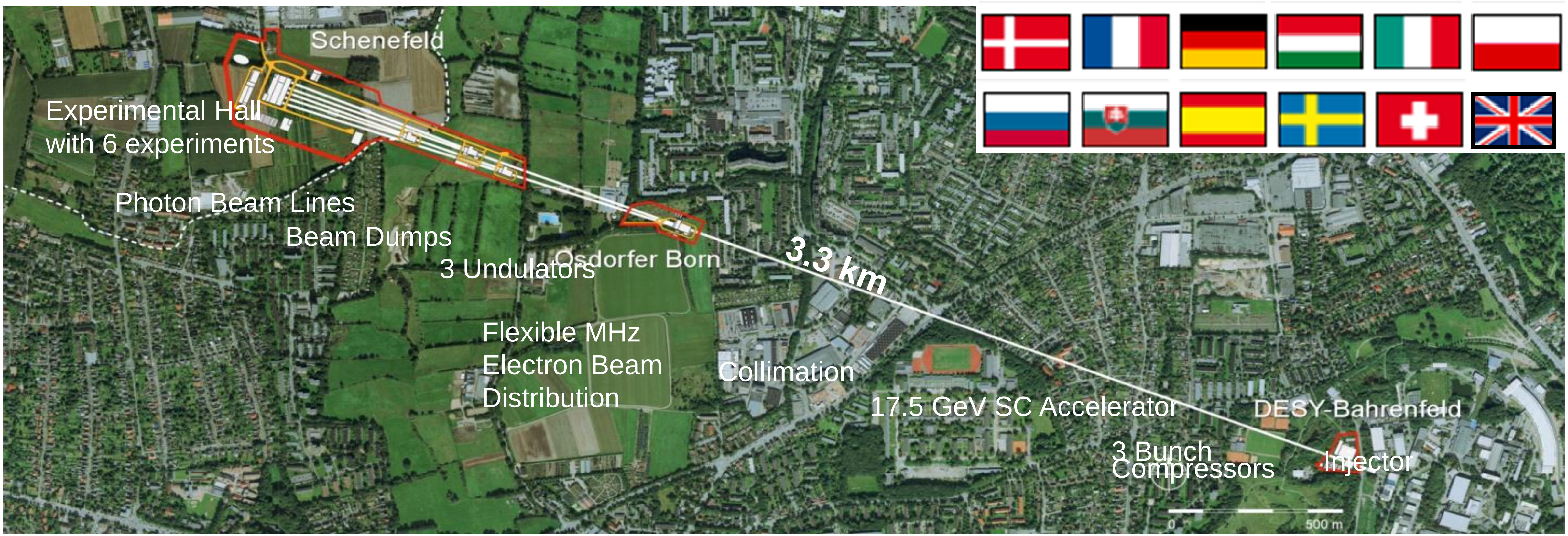
Deputy Project Leader European XFEL Construction Project

On behalf of the European XFEL Project and Operation Teams

UK Accelerator Institutes Seminar Series
25.11.2021

European XFEL at a Glance



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Project History

16 Years Preparation

7 Years Construction

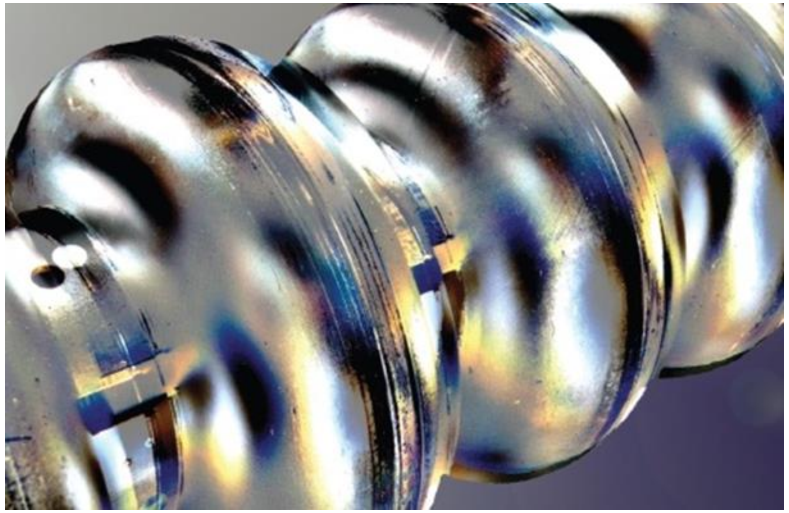
5 Years Operation

1993: Start of TESLA technology development

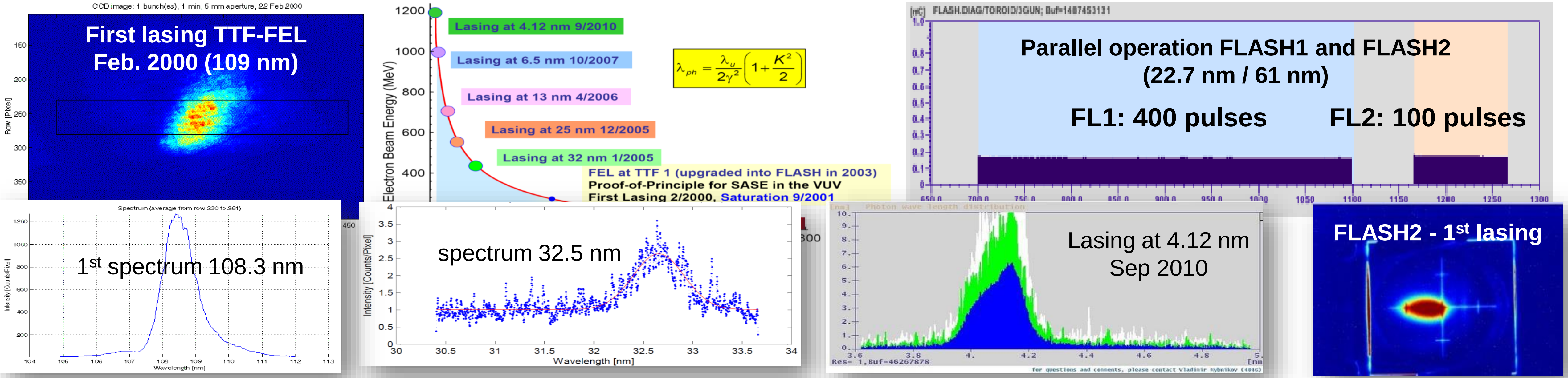
2000: First lasing at 109 nm at the TESLA Test Facility (TTF), now FLASH

Superconducting Technology

- Superconducting radiofrequency **(SRF) accelerators are a figurehead** of DESY‘s engagement in the design, construction and operation of accelerators for science.
- The used **TESLA technology** was developed since the early 90ies. **FLASH** is the first result of this R&D and can be seen as the prototype.
- The successful construction and commissioning of the European XFEL was the result of excellent cooperation within the DESY coordinated **Accelerator Consortium** consisting of 16 institutes.
- With the European XFEL the **fully successful technology transfer to industry** reached an important point. Other worldwide projects (LCLS-II, ESS, new SRF based FELs at e.g. SINAP, China) are profiting greatly from DESY efforts.



FLASH: the pioneering soft x-ray SASE FEL



First lasing TTF-FEL Feb. 2000 (109 nm)

First lasing FLASH Jan 2005 (32 nm)

Lasing at 4.12 nm Sep 2010

First lasing FLASH2 First parallel operation Aug 2014

First parallel User operation Apr 2016

1995 — 2000 — 2005 — 2010 — 2015 — 2020

Construction and operation TESLA Test Facility Linac (TTF)

Undulator installation

Operation TTF-FEL

Construction and commissioning FLASH

Upgrade to 1 GeV

Upgrade to 1.25 GeV

User operation FLASH1

Construction and commissioning FLASH2

User operation FLASH2

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2001: TESLA Linear Collider TDR with XFEL appendix

2006: European XFEL TDR

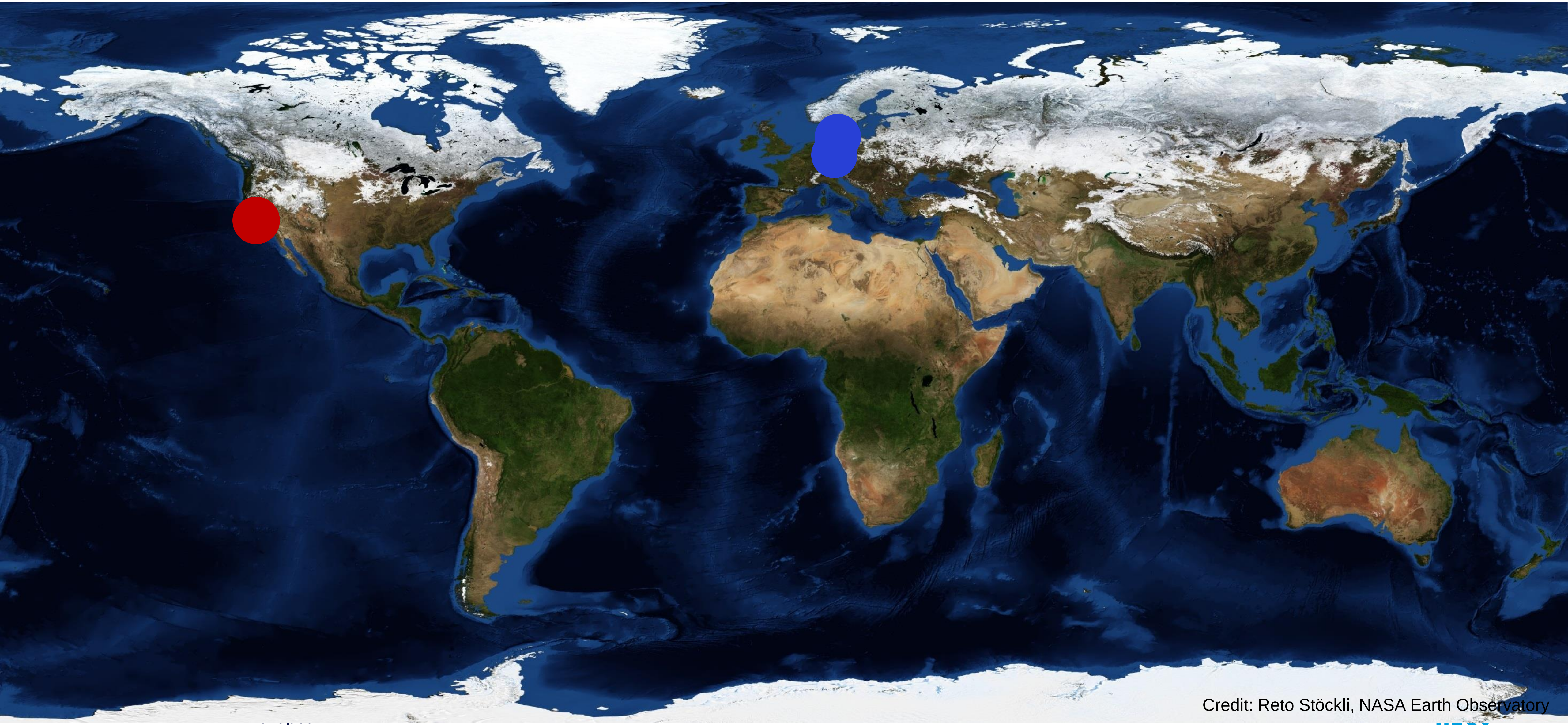
2009: Foundation of the European XFEL GmbH

Start of underground construction

2010: Formation of the Accelerator Consortium:

16 accelerator institutes under the coordination of DESY

X-Ray FELs around the world – 2009



Credit: Reto Stöckli, NASA Earth Observatory

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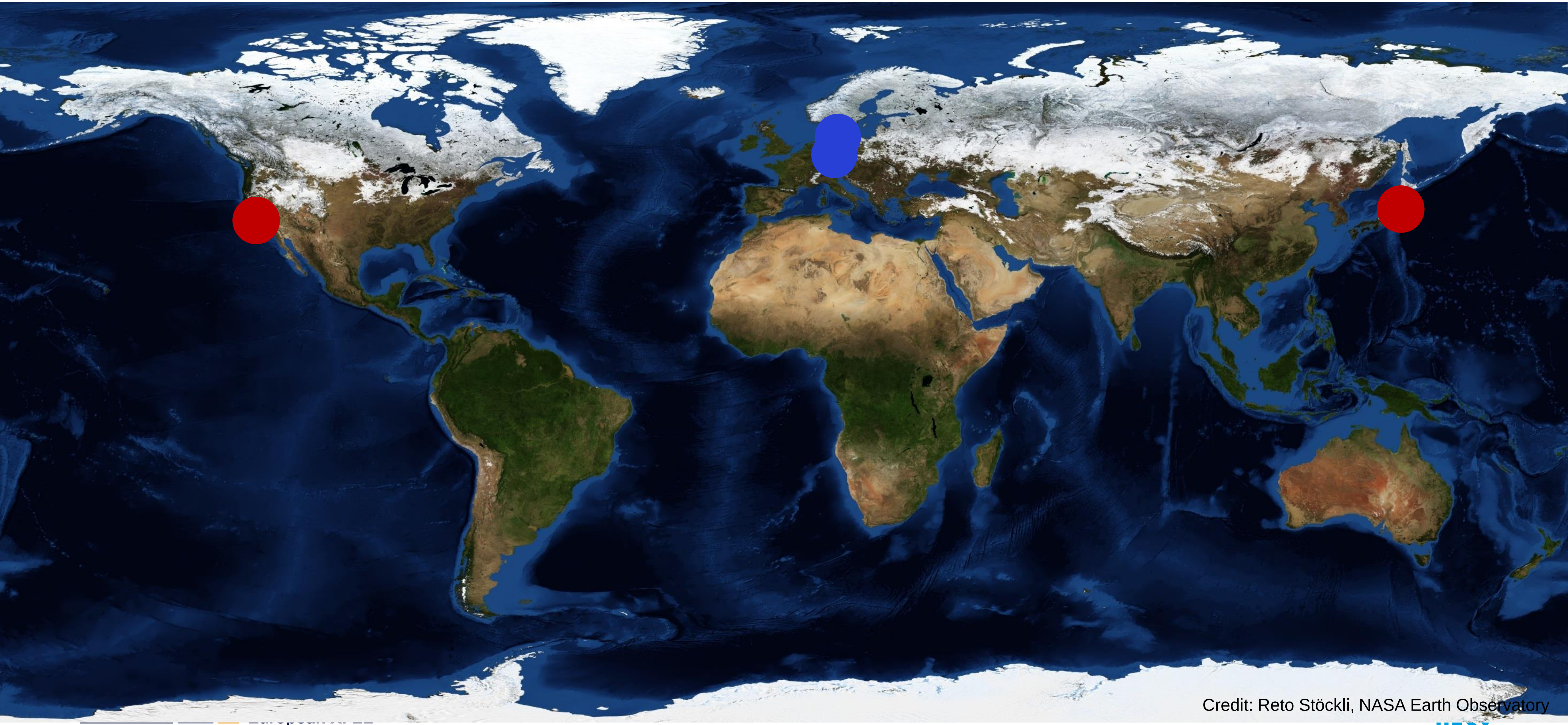
WTM

X-Ray FELs around the world – 2012



Credit: Reto Stöckli, NASA Earth Observatory

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Start of underground construction

2010: Formation of the Accelerator Consortium:

16 accelerator institutes under the coordination of DESY

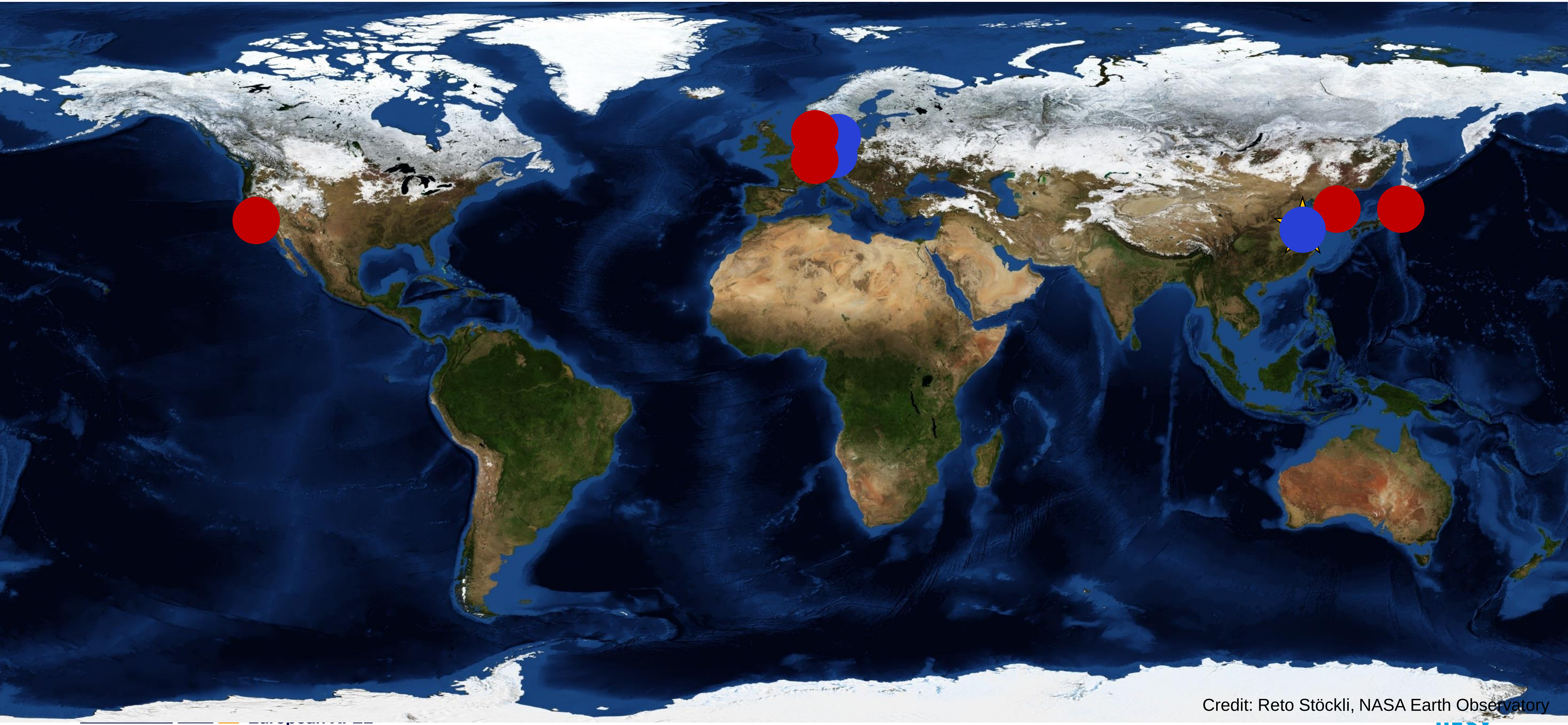
2012: End of tunnel construction, start of underground installation

2016: Finish of accelerator installation, start of commissioning

2017: First lasing SASE1, start of early user program

2018: First lasing SASE3 & SASE2, parallel operation of all 3 FELs

X-Ray FELs around the world – 2018



Credit: Reto Stöckli, NASA Earth Observatory

Back 2018/11/21

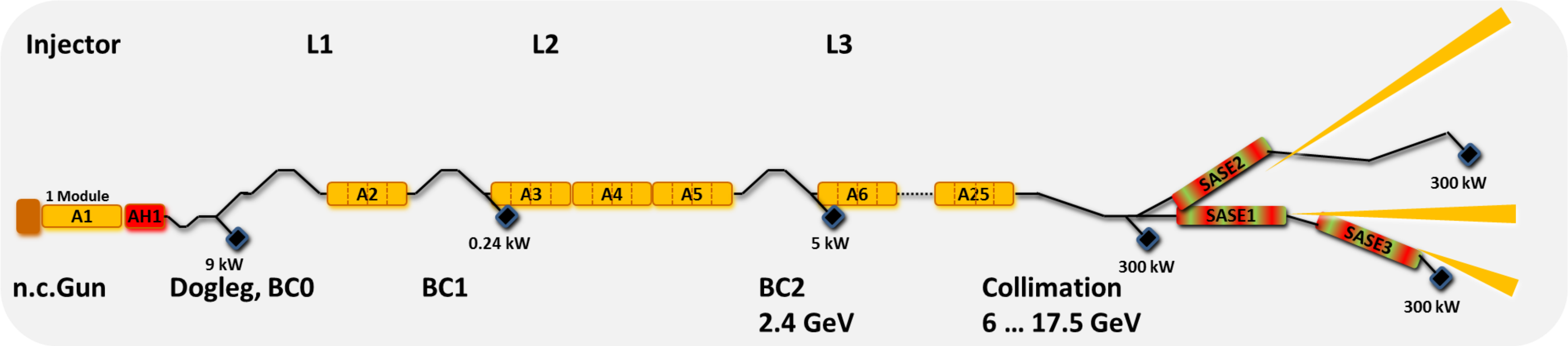
UK now European XFEL shareholder

UK RESEARCH AND INNOVATION JOIN THE EUROPEAN XFEL GMBH

On Wednesday 21 November, UK Research and Innovation (UKRI), an organization funded by the United Kingdom's Department for Business, Energy and Industrial Strategy, officially became a shareholder of European XFEL GmbH, the company that builds and runs the world's largest X-ray laser in Schenefeld near Hamburg.

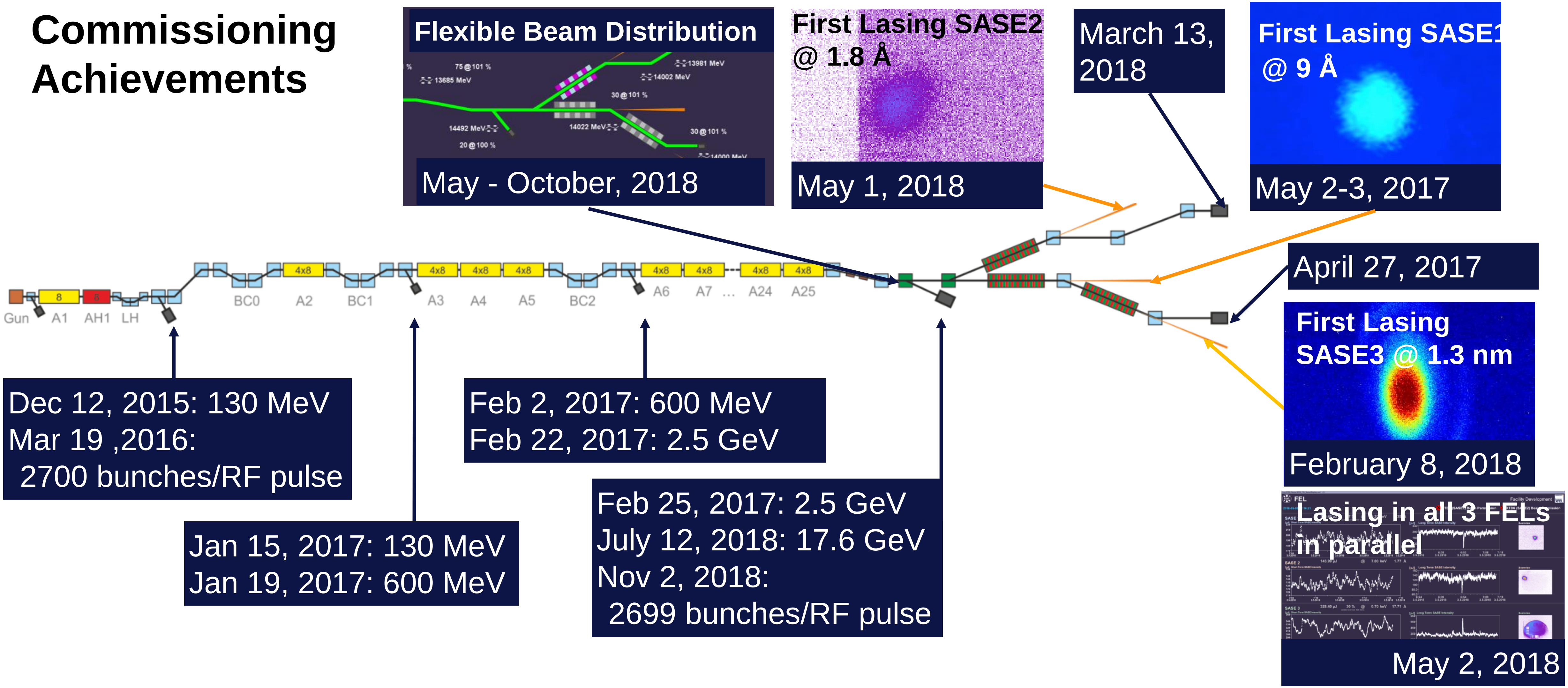
Back in March 2018, at the British Embassy in Berlin, UK representatives signed the

Superconducting Accelerators Offer Flexible Operation Parameters

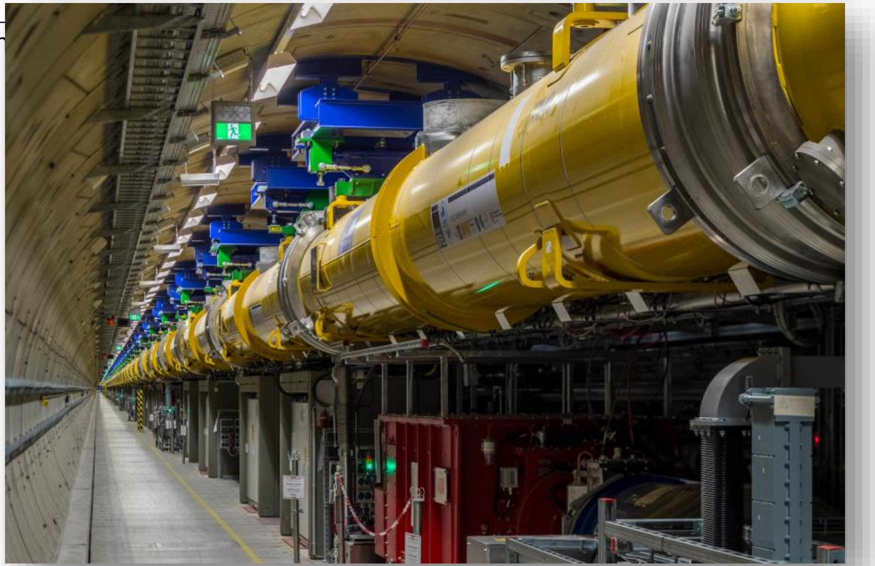


electron beam energy	8 / 12.5 / 14 / 17.5 GeV
macro pulse repetition rate	10 Hz
RF pulse length (flat top)	600 µs
# of bunches/second	27,000
bunch charge	0.02 – 1 nC
electron bunch length after compression (FWHM)	2 – 180 fs
normalized slice emittance	0.4 mm mrad @ 0.25 nC
simultaneously operated SASE undulators	3

Commissioning Achievements




Worlds largest superconducting linear accelerator

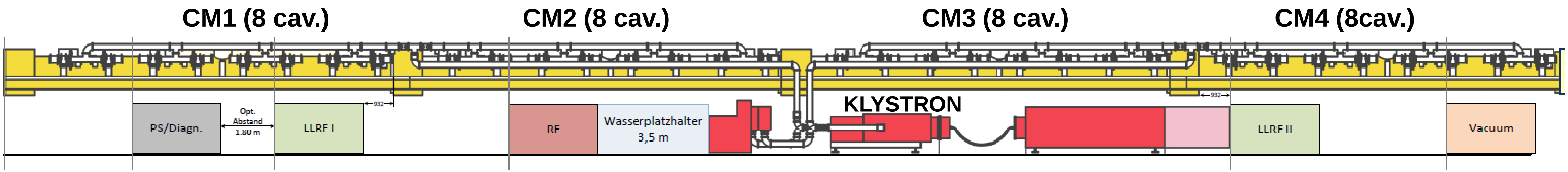


The European XFEL Accelerator (RF station)

One RF station comprises*:

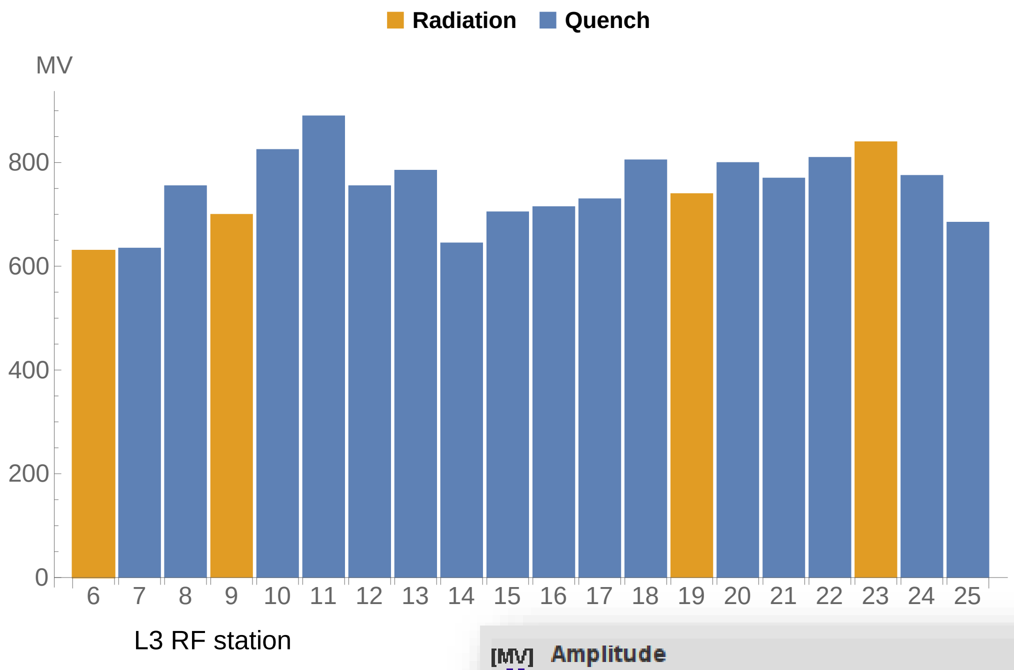
- 1× 10 MW **klystron**
- 32× TESLA-type 1,3-GHz **cavities** housed in 4 **cryomodules**
- 32× motorized power **couplers**
- 32× motorized **tuners**
- 64× **piezo** (actuator actuator / sensor)
- 36× motorized phase shifters (1/ cav + 1/ cryomodule)
- 100+ LLRF channels (probe, forward, reflected)

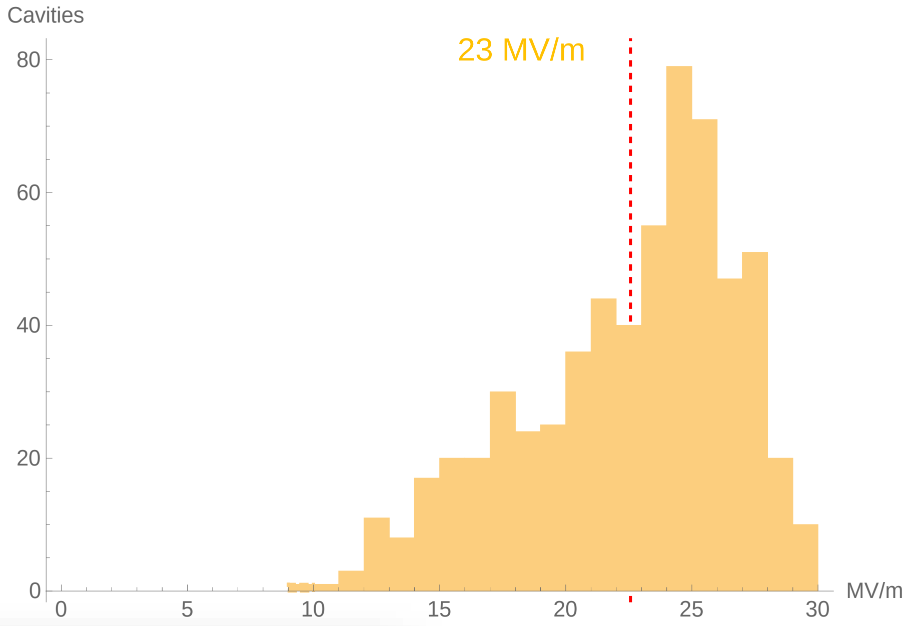
25 RF Stations
- **25 Klystrons**
- **97 Cryomodules**
- **776 1.3 GHz cavities**

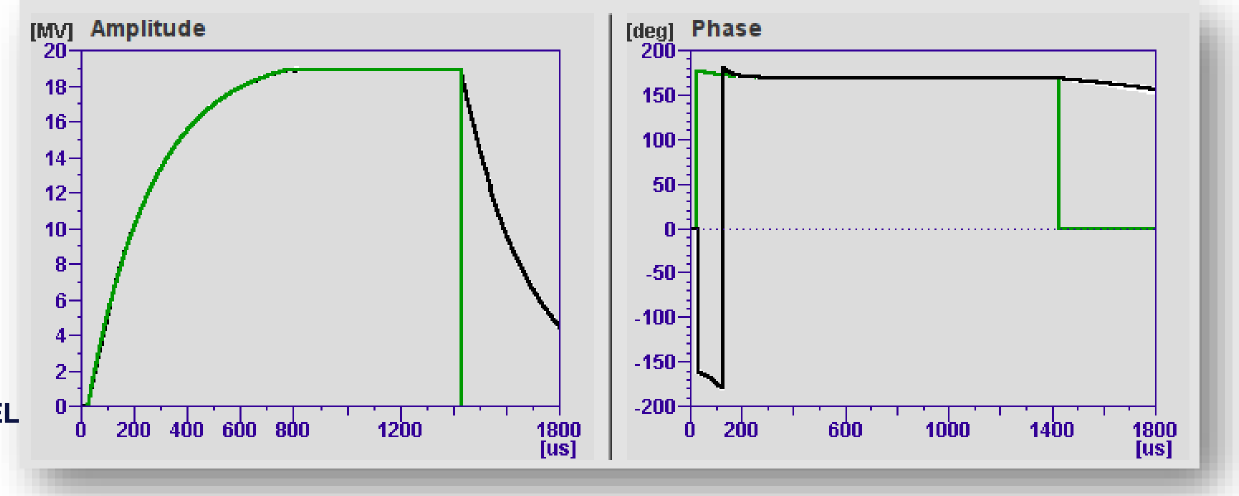


* Exception: A1 (injector) 1 CM

It’s all about VOLTS







SRF LINAC operations

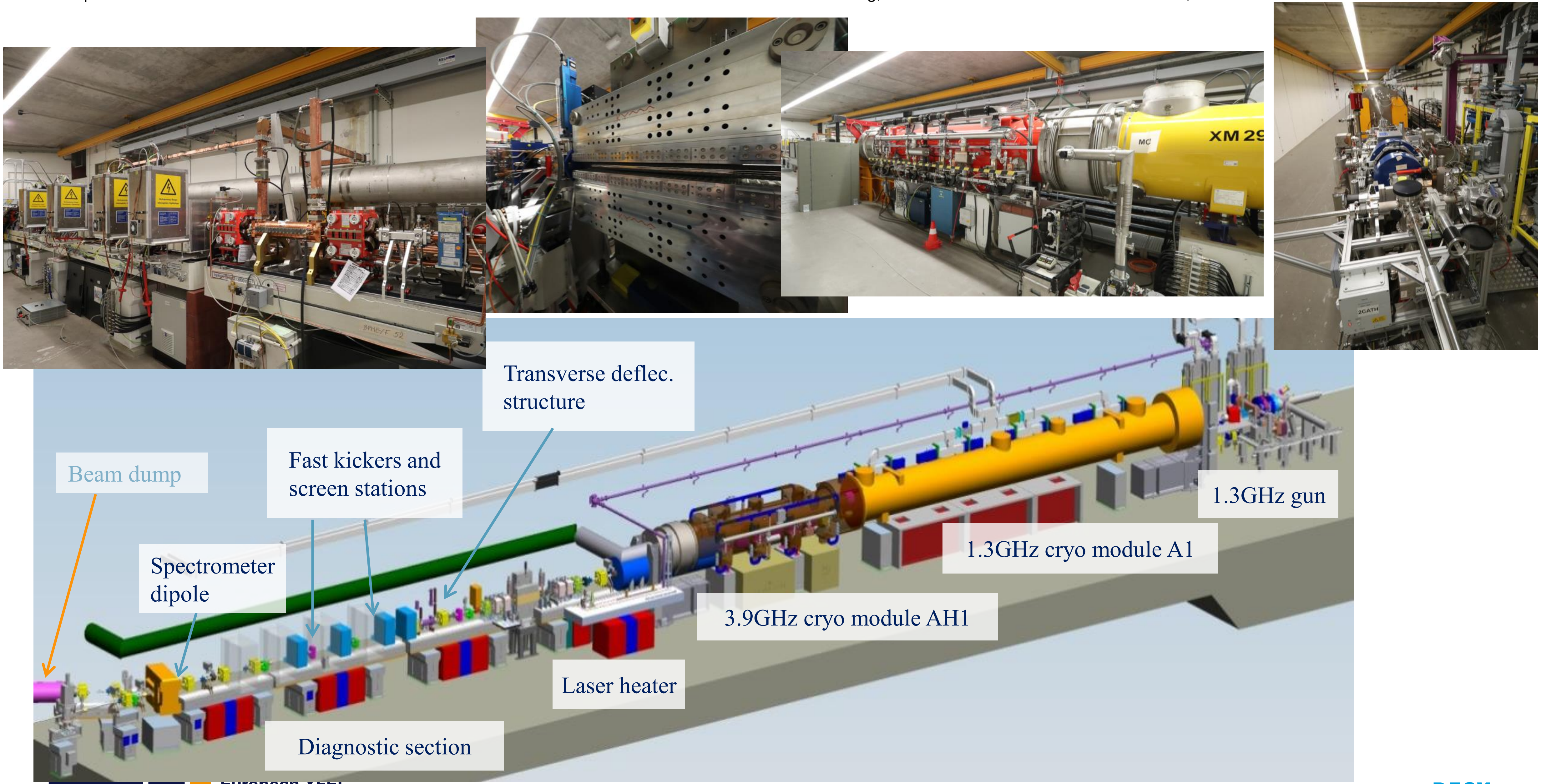
Maintaining HA remains operations focus

- SRF linac maintains High Availability (RF)
- 16.3 GeV High Voltage (higher radiation) operation not significantly worse
- Weekly Routine Maintenance to check / reset (mostly) LLRF state
  - Calibration
  - Phase alignment
  - System issues etc
- BUT! HA doesn't come for free
  - Diligence!
  - Big team effort 'behind the scenes'
  - LinacOps meetings (ops review)

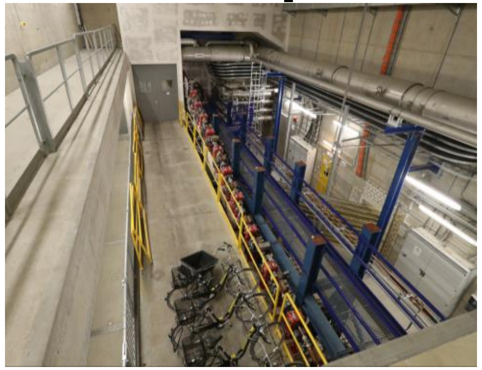
what you see

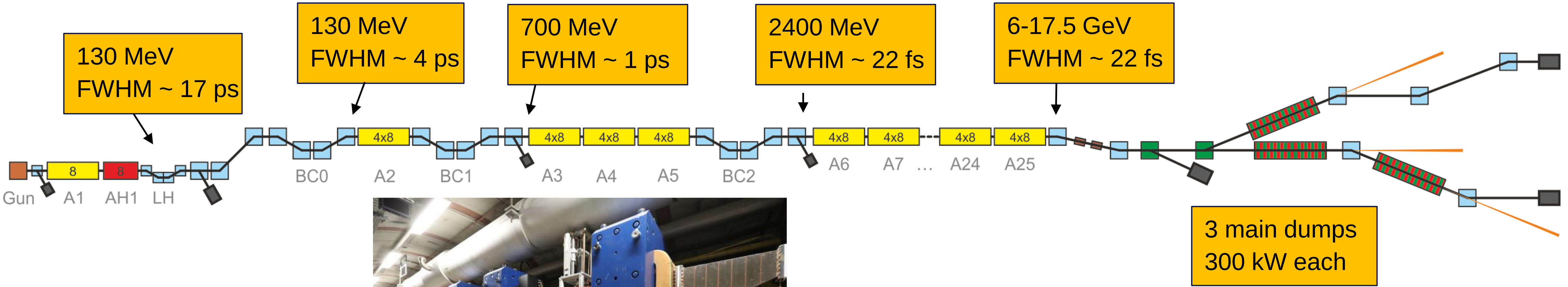
what's going on behind the scenes

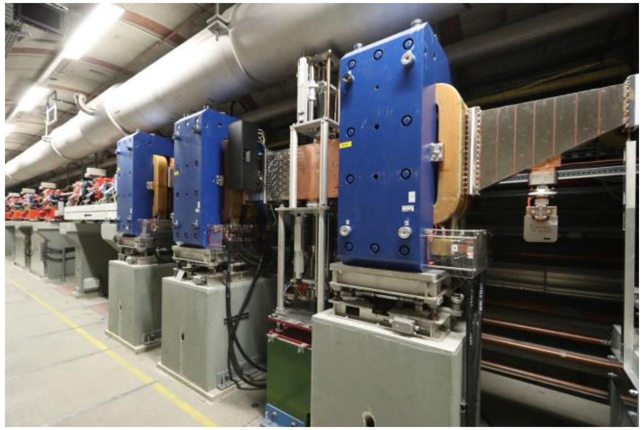
EuXFEL - Injector


Transverse deflec. structure
Fast kickers and screen stations
Beam dump
Spectrometer dipole
1.3GHz gun
1.3GHz cryo module A1
3.9GHz cryo module AH1
Laser heater
Diagnostic section

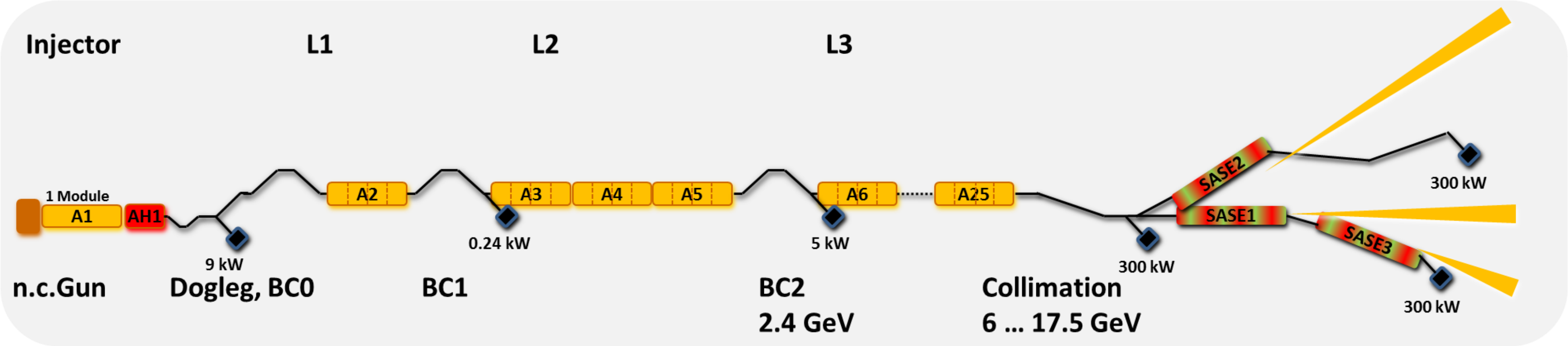
3 Bunch Compression Stages







Superconducting Accelerators Offer Flexible Operation Parameters



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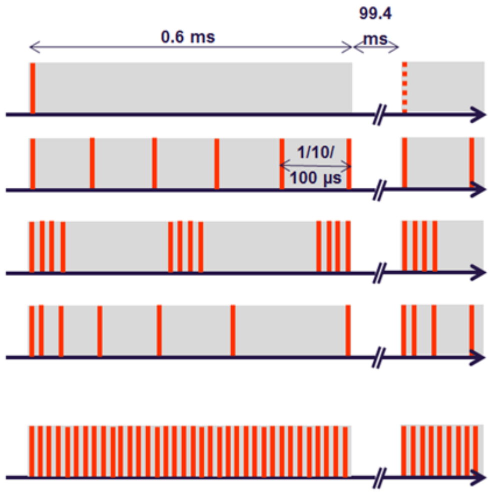
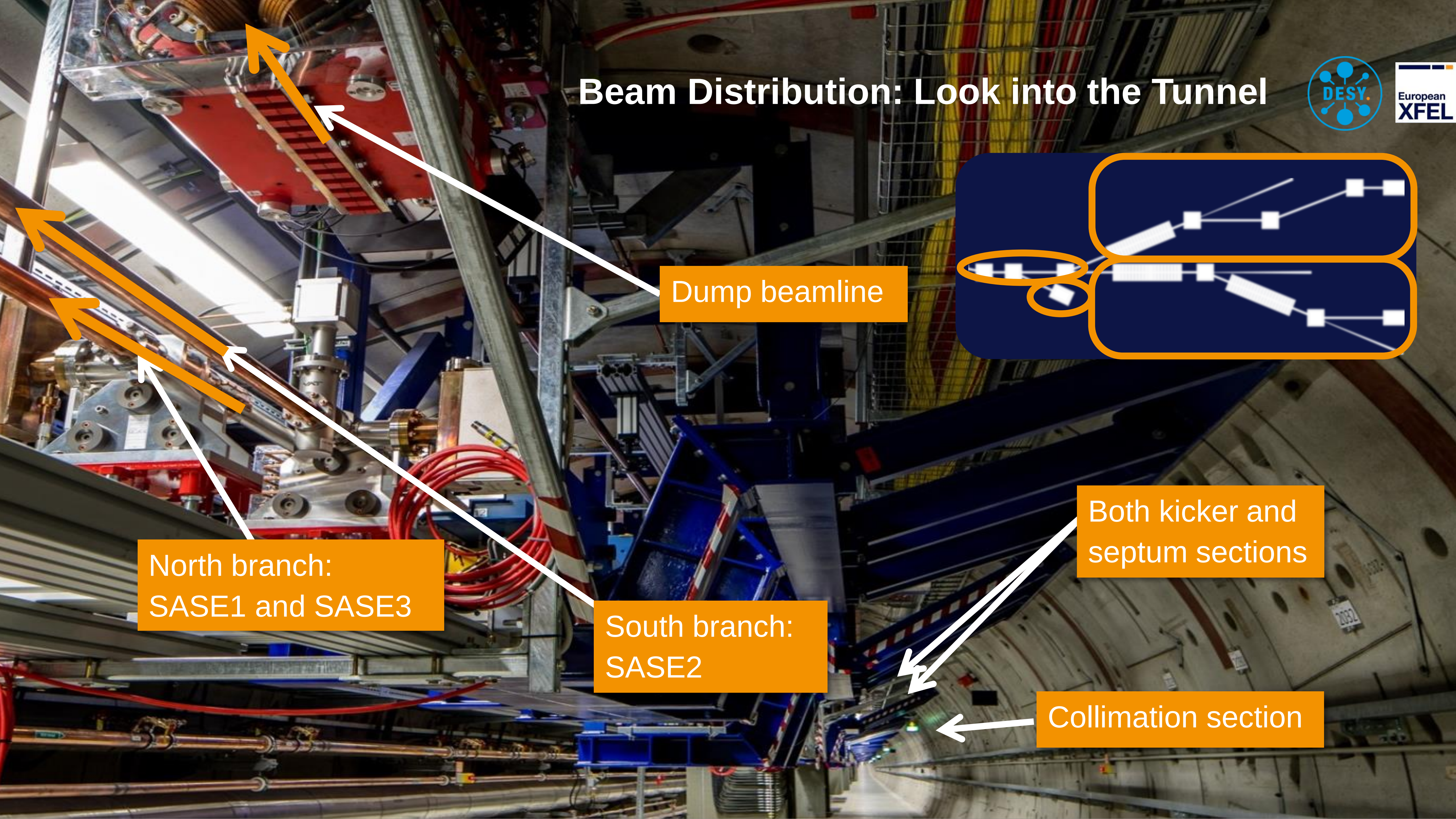
simultaneously operated SASE undulators	3

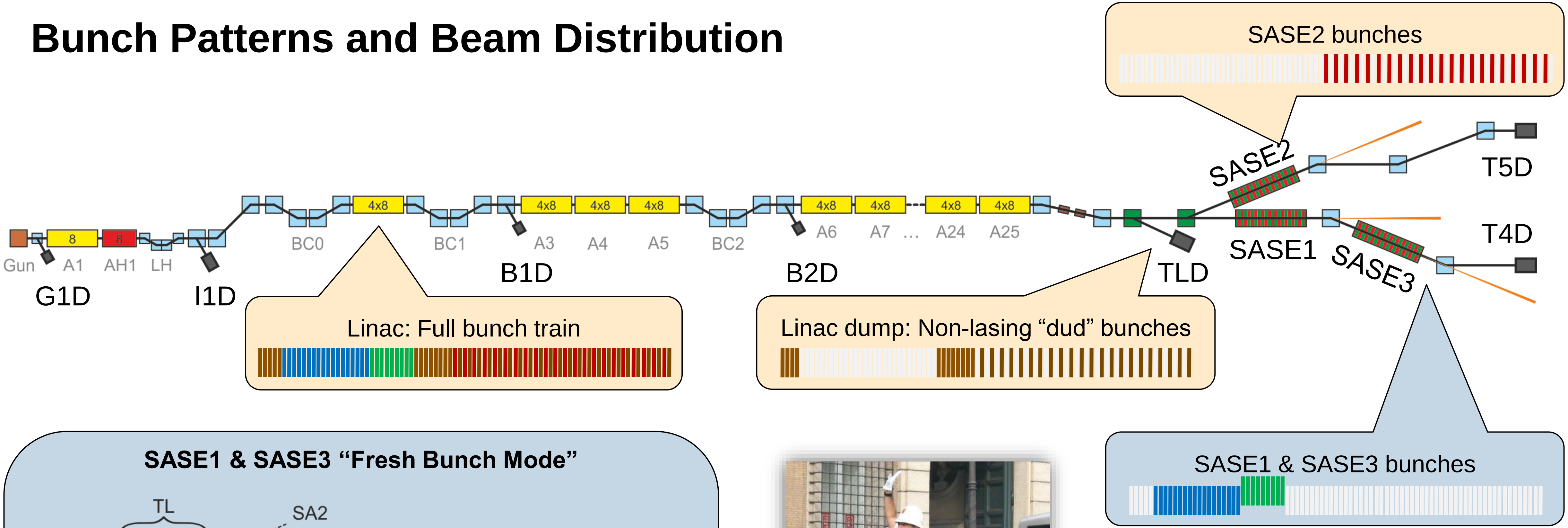
- Long RF pulses offer **high flexibility** wrt. time structure (already in pulsed mode).
- Parallel operation of undulator lines.

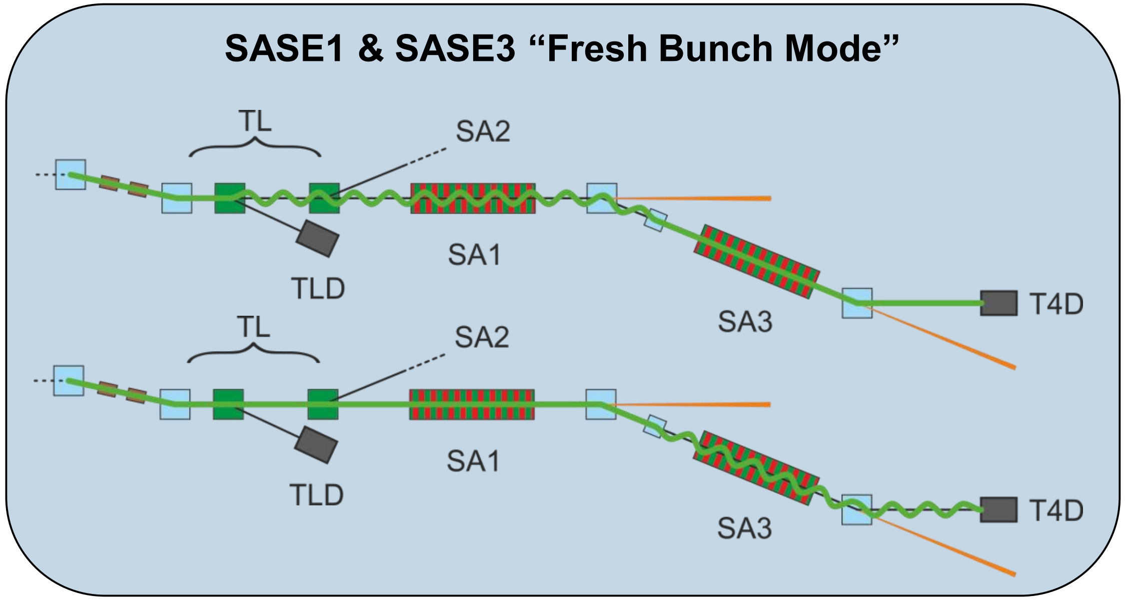


Beam Distribution: Look into the Tunnel

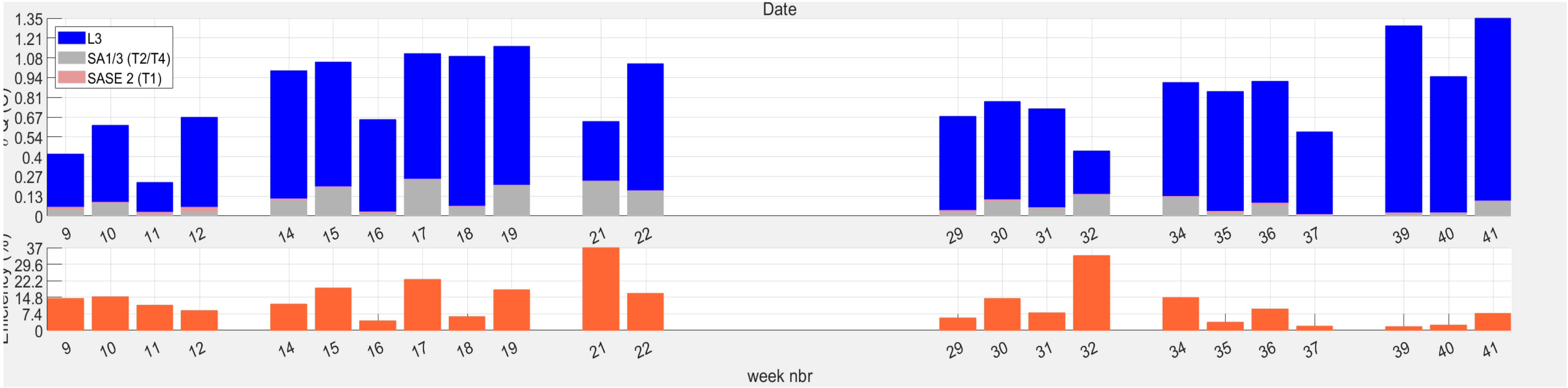


Bunch Patterns and Beam Distribution



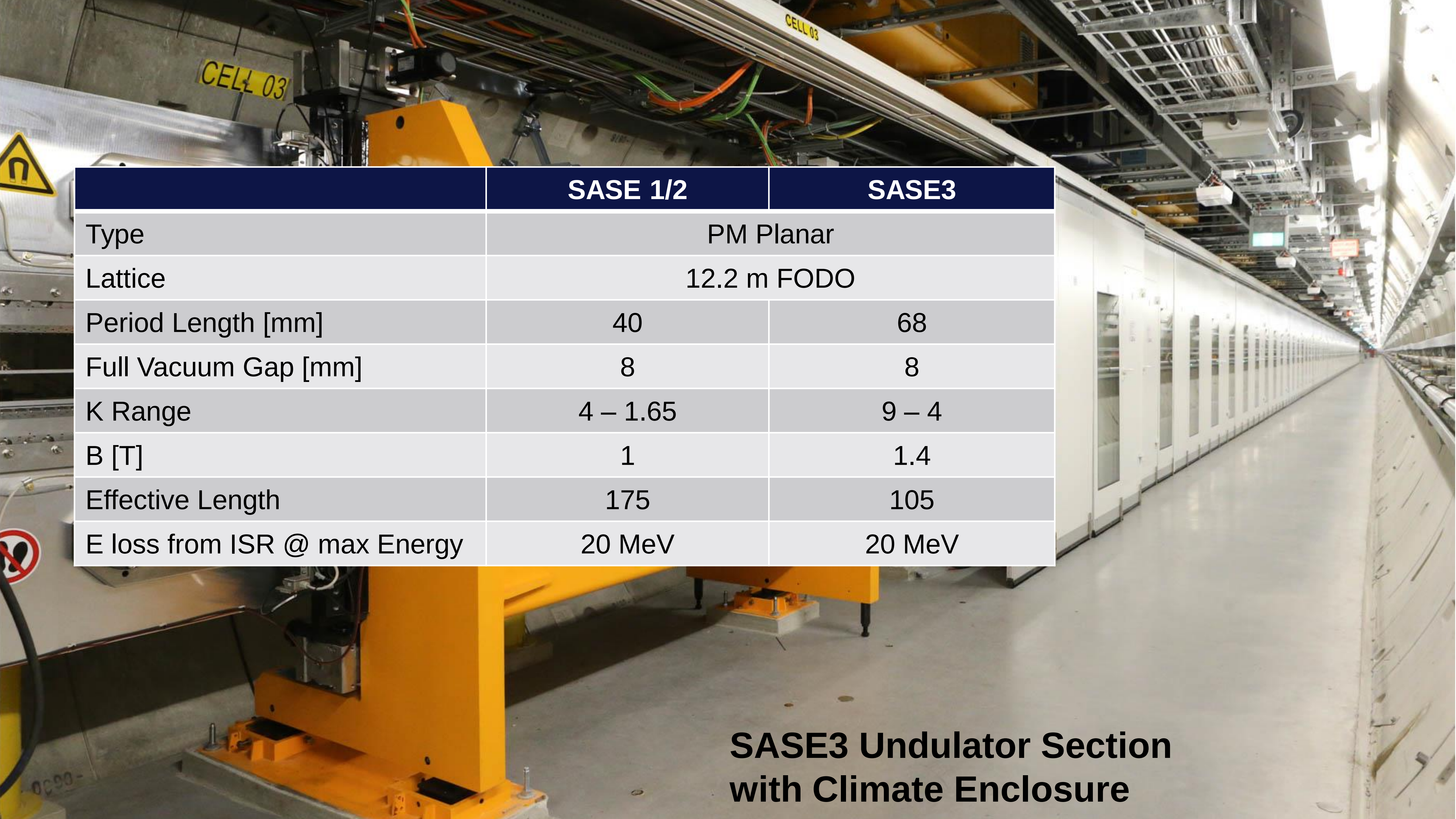


Usage of electron bunches in delivery weeks



- On average only about 15% of the available electrons are used for photon production
- Has several reasons, have to tackle where possible
  - Photon beam intensity safety limits
  - Detector performance
  - Sample delivery
  - Set-up & alignment of experiments
  - …

SASE3 Undulator Section with Climate Enclosure

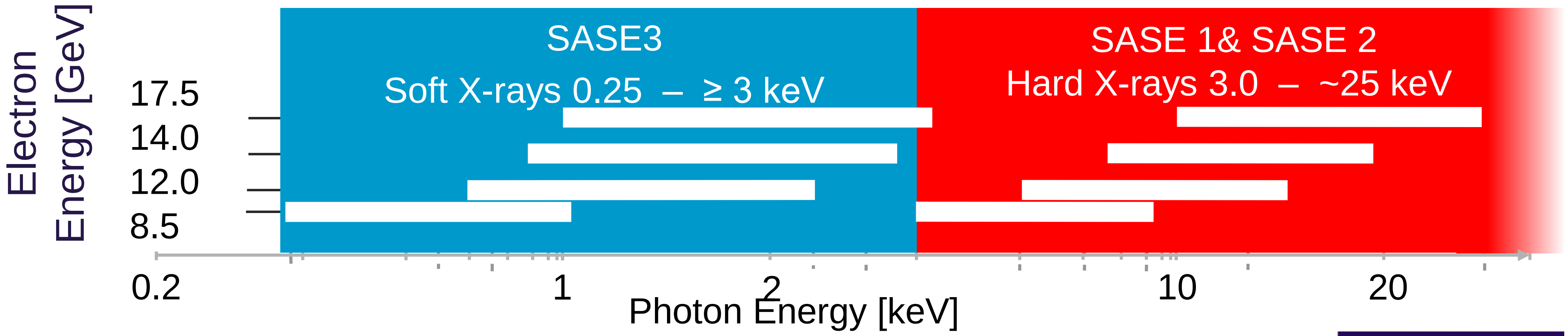


	SASE 1/2	SASE3
Type	PM Planar	
Lattice	12.2 m FODO	
Period Length [mm]	40	68
Full Vacuum Gap [mm]	8	8
K Range	4 – 1.65	9 – 4
B [T]	1	1.4
Effective Length	175	105
E loss from ISR @ max Energy	20 MeV	20 MeV

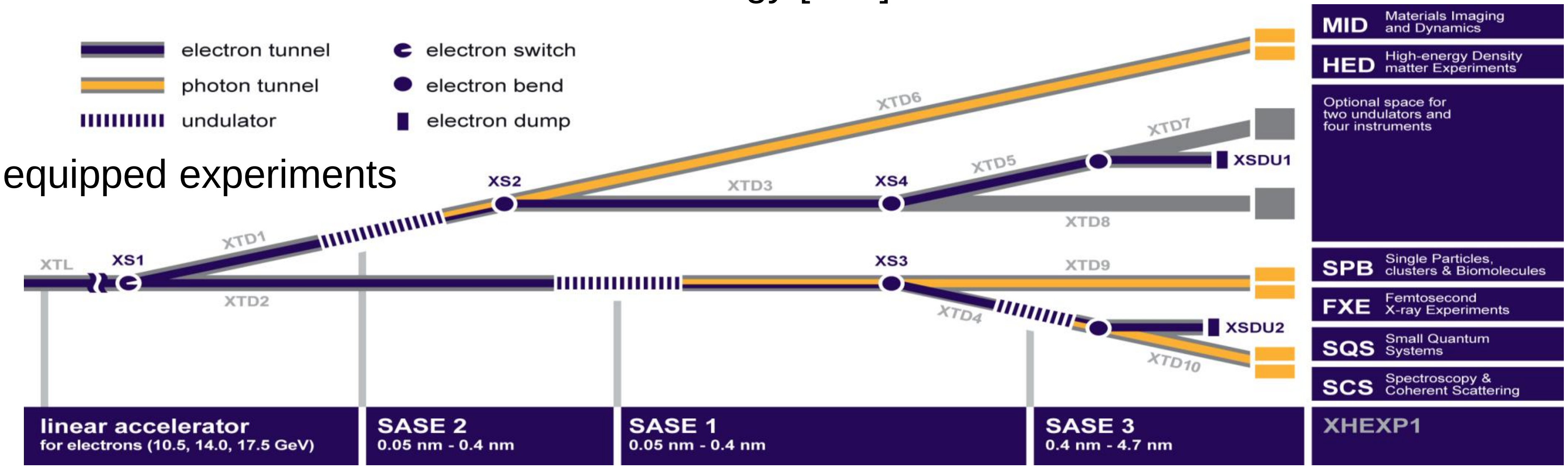
SASE3 Undulator Section with Climate Enclosure

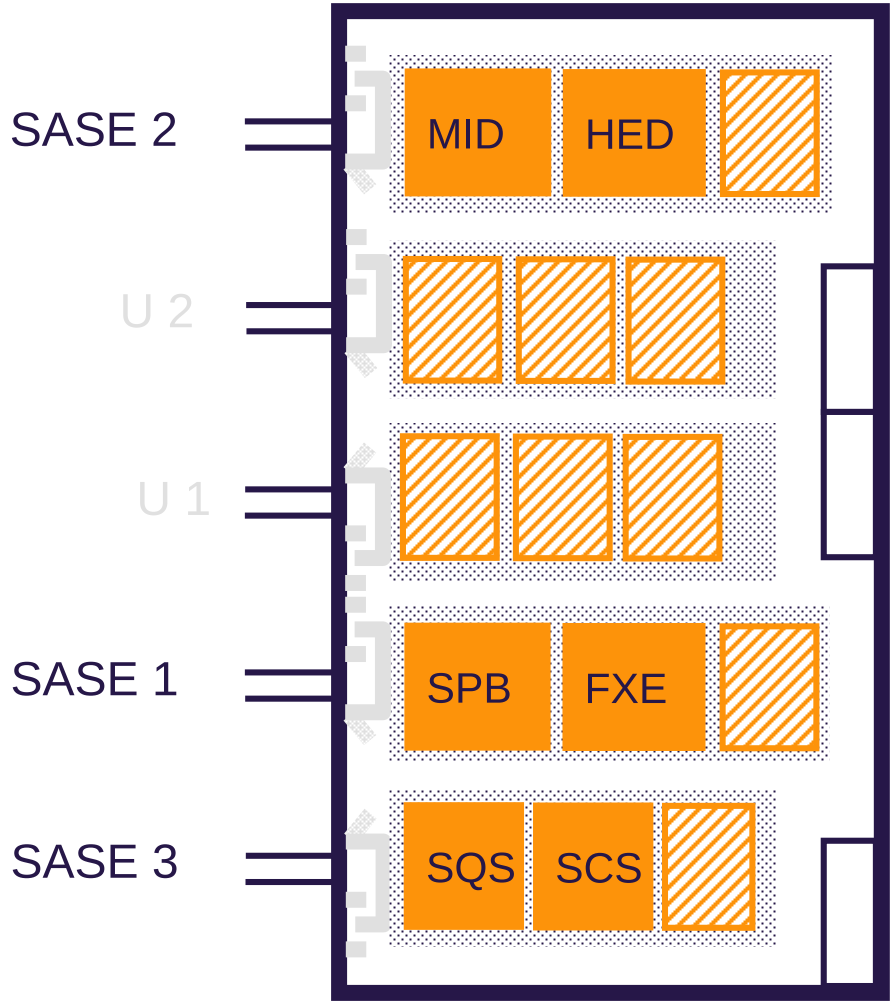
Covers photon energies from 0.25 keV to 25 keV

Three variable gap undulators for hard and soft X-rays operated in parallel

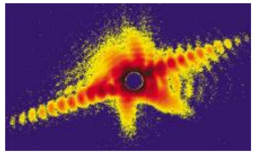


Initially 6 equipped experiments

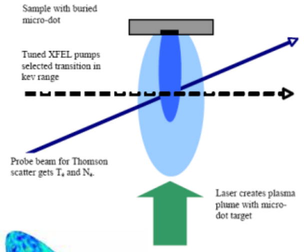




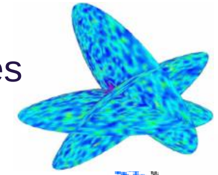
MID Materials Imaging & Dynamics



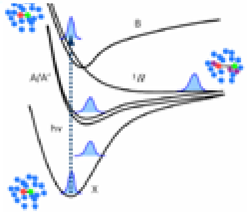
HED High Energy Density Science



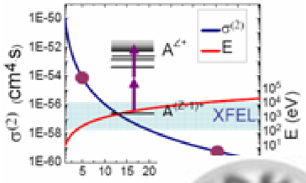
SPB Single Particle & Biomolecules



FXE Femtosecond X-ray Experiments



SQS Small Quantum Systems



SCS Spectroscopy & Coherent Scattering

More about experiments: http://www.xfel.eu

Undulator Operation: Standard FEL Tuning Procedure

After Shut-Downs (twice per year)

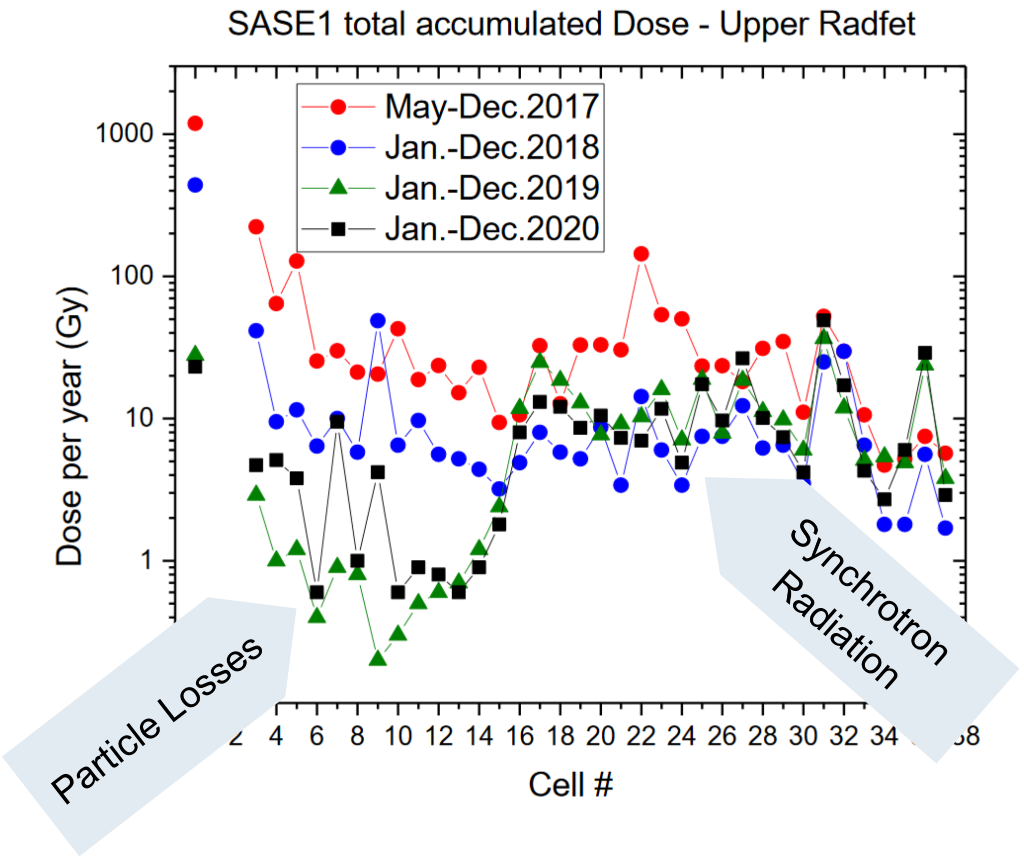
1) Beam Based Alignment
2) Adjust Undulator mid-plain to BBA Orbit using (K-Mono)
3) Measure and Correct Undulator K-Offsets (K-Mono)
4) Measure and apply Wakefield compensating linear taper

Each delivery run (weekly)

1) Optimize Accelerator: Emittance, Dispersion, Compression, Orbit, Matching, Laser Heater
2) Establish initial SASE level
3) Fine tune linear taper in gain regime and optimize quadratic taper by maximizing power
4) Empirical fine tuning (for power) at each undulator gap setting, wavelength tune-ability suffers but intensity gains
5) Adjust pointing

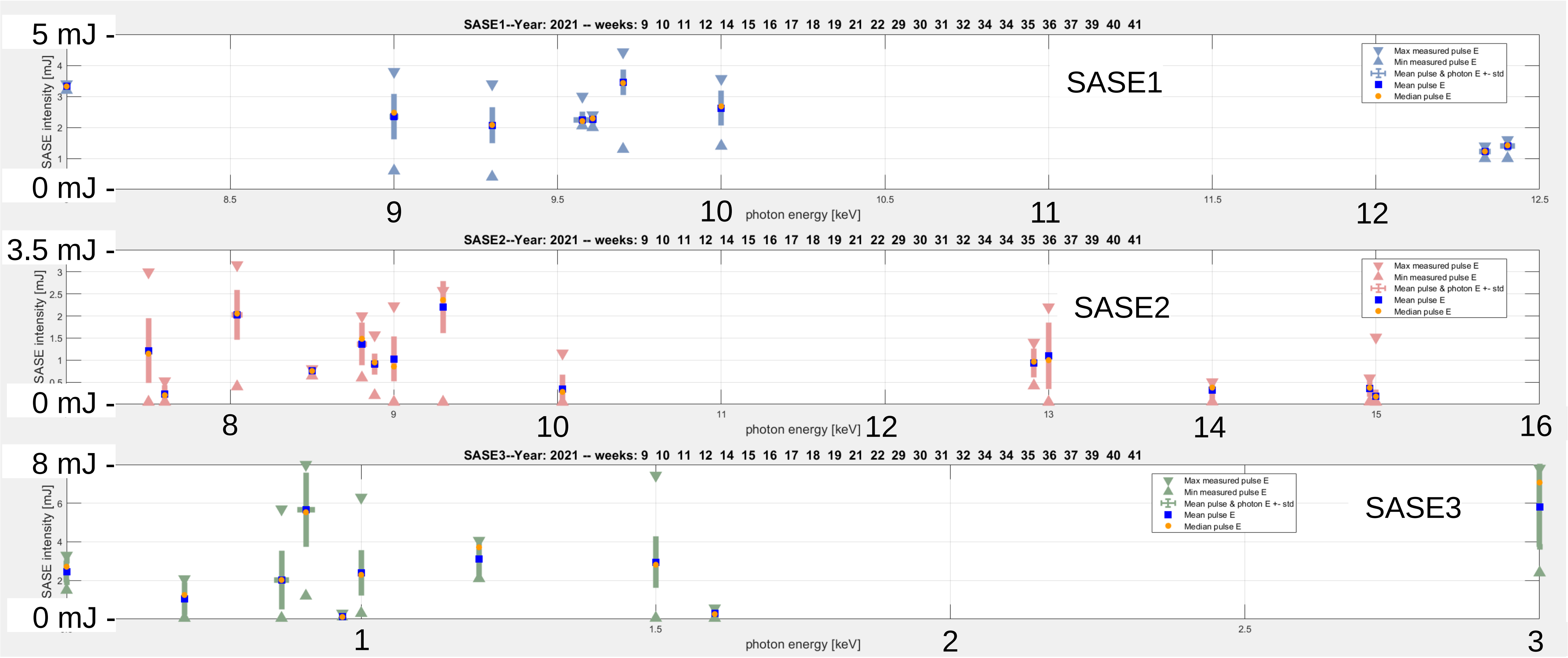
Losses – are not a problem

Dominated by synchrotron radiation
Particle losses account for a few Gy/year
PM damage (1e-4 field drop) expected at about 200 Gy

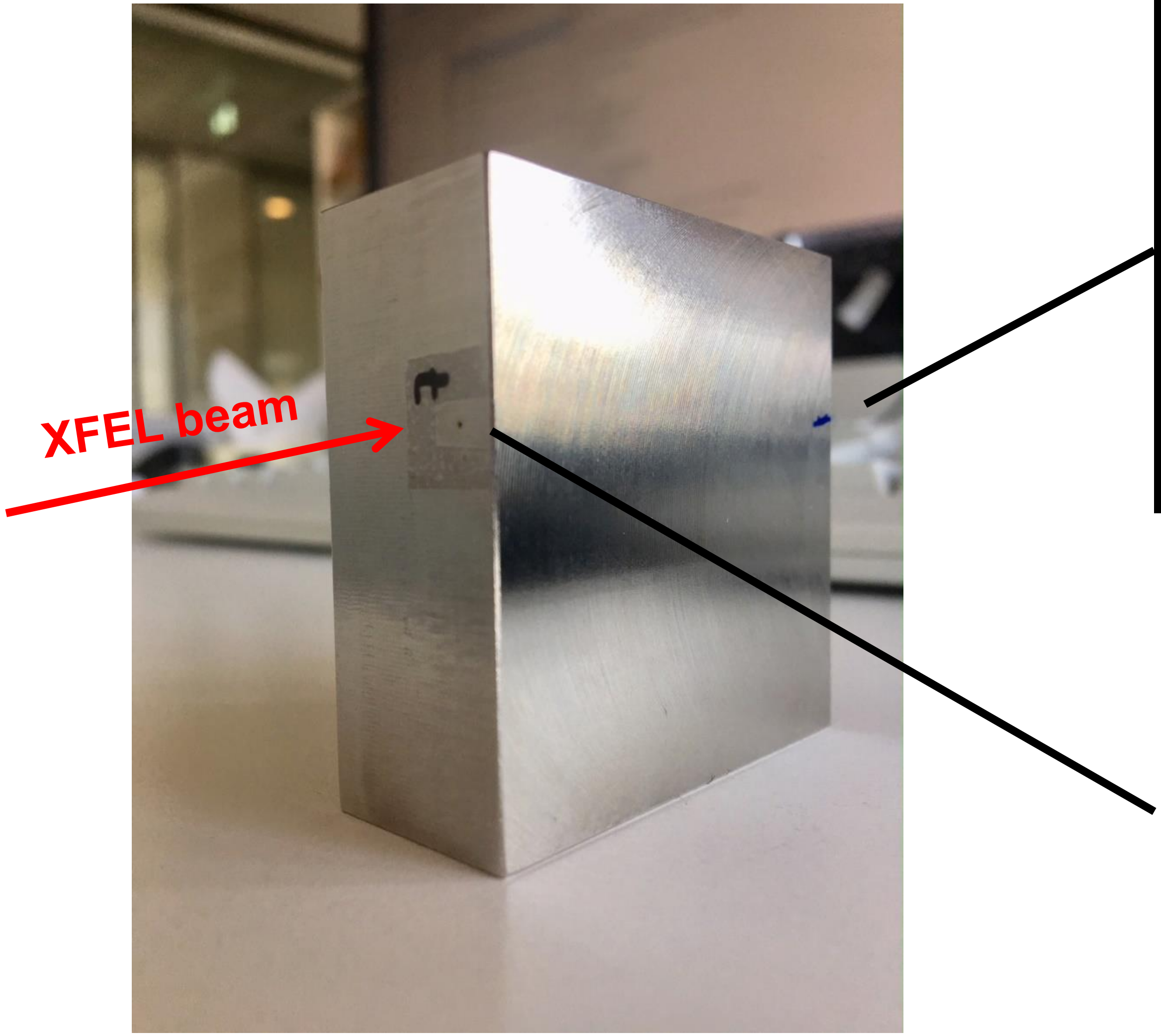


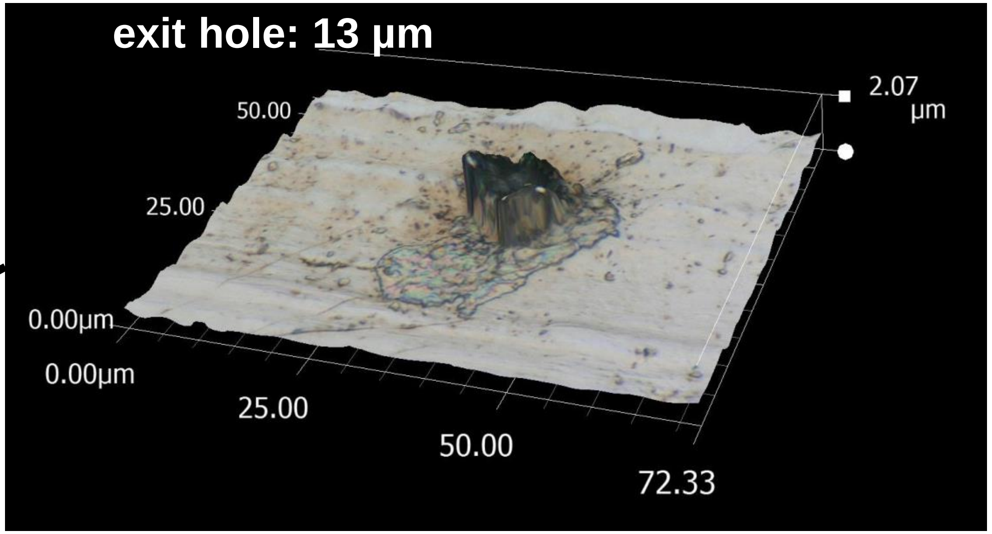
SASE performance in 2021

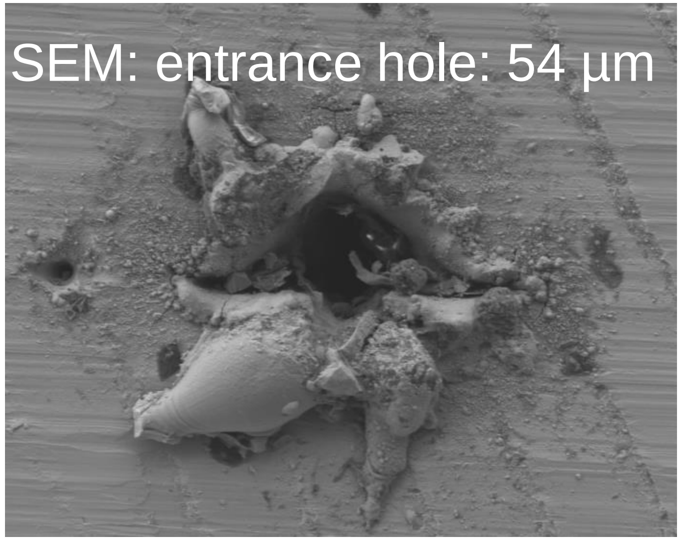
SASE pulse energy in mJ depending on photon energy in keV



Powerfull Beams: Drilling with XFEL Beam Through 50 mm of Steel in 26 Seconds



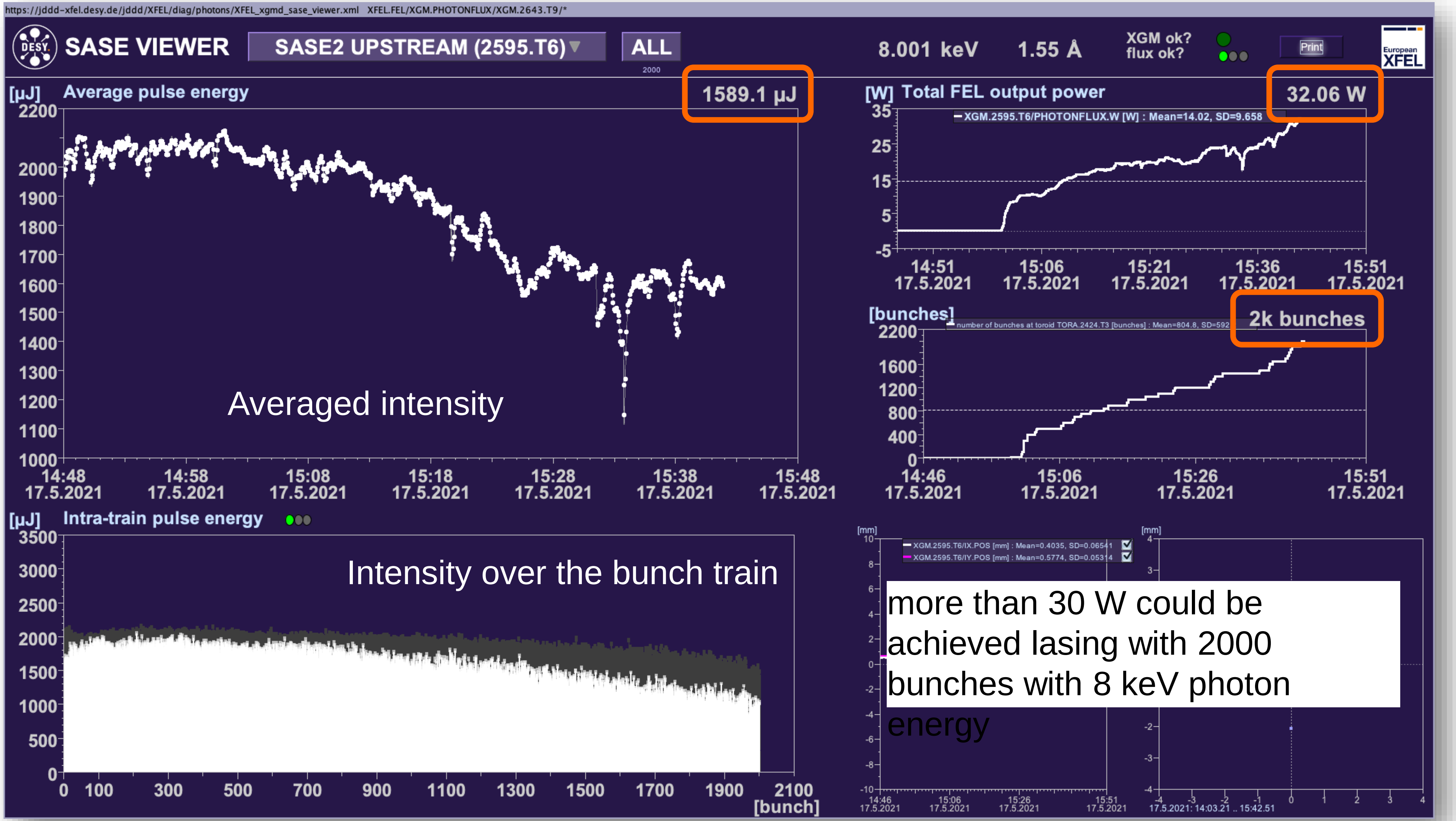




FXE instrument
9.3 keV
1 mJ/pulse
20 µm beam size
1500 pulses/sec

High power test in SASE2

Setting new records



more than 30 W could be achieved lasing with 2000 bunches with 8 keV photon energy

High power limits

Expanding the allowed operation envelope

- FEL beam can drill through everything (if focused)
- Active beam stop system deployed at beam stops and experiments
- Power limits increased to 40 W (from 0.8 to 8 W)

XFEL Operation Constraints September 8, 2021

		Limit	Restriction
1	SASE 1		
1.1	Photon power	≤ 40 W	The maximum photon power may not exceed 40 W.
2	SASE 2		
2.1	Photon power	≤ 40 W	The maximum photon power may not exceed 40 W.
3	SASE 3		
3.1	Photon power	≤ 40 W	The maximum photon power may not exceed 40 W.

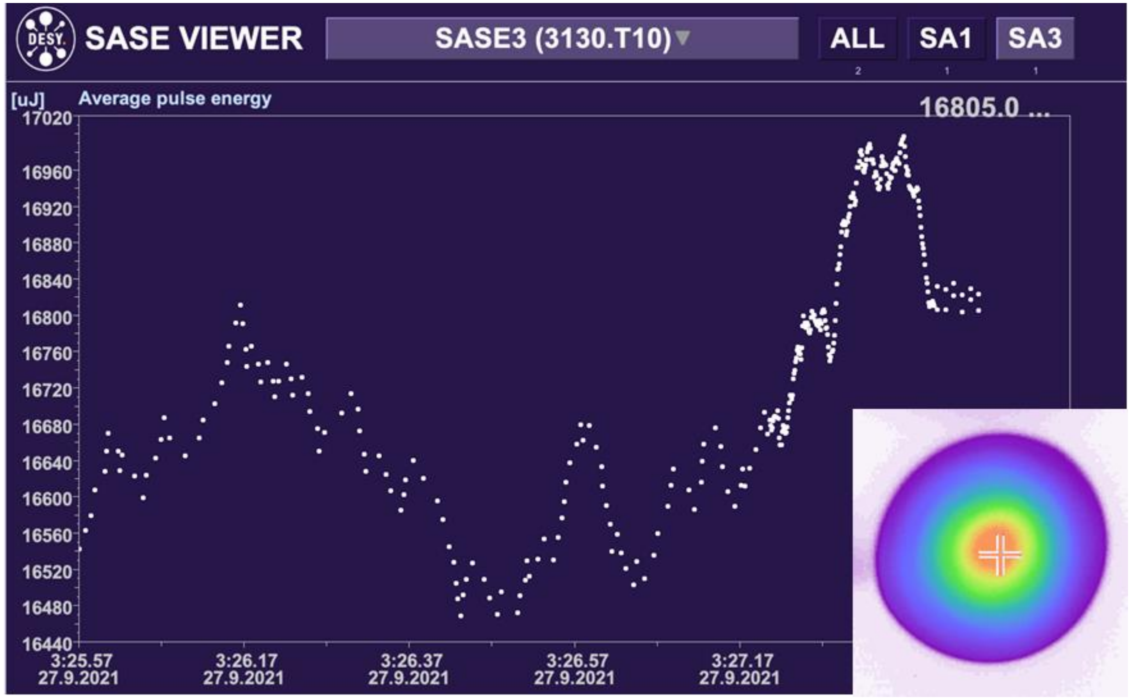
XFEL Operation Constraints August 24, 2020

		Limit	Restriction	To be ensured by:
1	General			
1.1	Number of bunches	≤ 400	The maximum number of bunches per pulstrain may not exceed 400 in total SASE 1+3 and 400 in SASE 2	organizationally
1.2	Undulators open	no limit	When all undulators are open (SASE 1, 2 and 3), no limits on the number of bunches apply	organizationally
2	SASE 1			
2.1	Photon energy	≥ 6 keV	The photon energy must be greater or equal 6 keV	organizationally
2.2a	Photon power (6 keV - 8.1 keV)	< 4 W	For photon energies between 6 keV and 8.1 keV the maximum photon power may not exceed 4 W	organizationally
2.2b	Photon power (> 8.1 keV)	< 8 W	For photon energies greater than 8.1 keV the maximum photon power may not exceed 8 W	organizationally
3	SASE 3			
3.1a	Photon power (≤ 1.2 keV)	< 30 W	The maximum photon power may not exceed 30 W at photon energies less or equal than 1.2 keV	organizationally
3.1b	Photon power (1.2 keV - <2.5 keV)	< 9 W	For photon energies between 1.2 keV and less than 2.5 keV the maximum photon power may not exceed 9 W	organizationally
3.1c	Photon power (≥ 2.5 keV)	< 5 W	For photon energies greater or equal 2.5 keV the maximum photon power may not exceed 5 W	organizationally
3.2	Pre-absorber		The photon beam must not hit the SQS or SCS safety absorbers (diamond/B4C pre-absorber in the beam or respective safety absorber open)	technical measures (SEPS interlock)
4	SASE 2			
4.1	Photon energy	≥ 5.8 keV	The photon energy must be greater or equal 5.8 keV	organizationally
4.2a	Photon power (5.8 keV - <8.8 keV)	< 0.8 W	For photon energies between 5.8 keV and less than 8.8 keV the maximum photon power may not exceed 0.8 W	organizationally
4.2b	Photon power (≥ 8.8 keV)	< 3 W	For photon energies greater or equal 8.8 keV the maximum photon power may not exceed 3 W	organizationally
4.3	Pulse energy	< 3 mJ	The maximum photon pulse energy may not exceed 3 mJ	organizationally
4.4a	CRL XTD 1	out	If the absorber TD6 in XTD 1 is not open the CRL 1 arms 7, 8 and 9 must be safely driven out of the beam	technical measures (SEPS interlock)
4.4b	CRL XTD 6	out	The following lenses of CRLs in SASE2/XTD6 must be safely driven out of the beam: CRL MID: Disable arms 7+8+9 CRL HED: Disable arms 4+5+6+7+8+9+10 *(numbering from arm 1, even when empty)*	technical measures
4.5	Beamshutter 6.1.1/6.1.2	open	The absorber/beamshutter 6.1.1 and 6.1.2 in the HED optics and HED experiment hutch must be blocked open	technical measures
4.6	Beamshutter 6.2.1	open	The absorber/beamshutter 6.2.1 in the MID optics hutch must be blocked open	technical measures

High power limits

Pushing the boundaries further – Record intensities at SASE3: 17 mJ at 1 keV

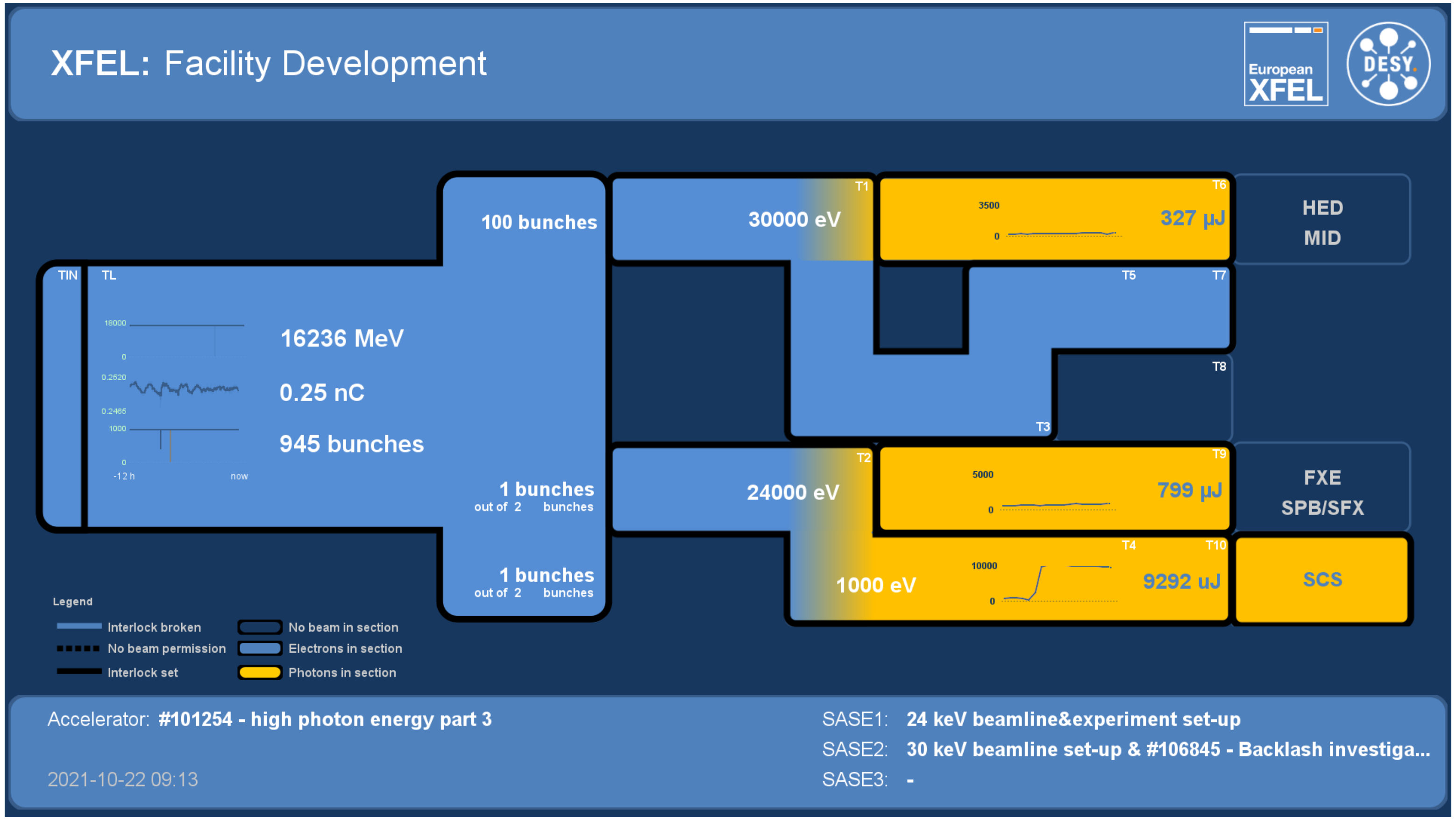
- Combination of
  - Unique compression
  - Undulator alignment
  - High electron energy



Higher Photon Energies

Increased Photon Energy Ranges- already incorporated into 2022-II Call

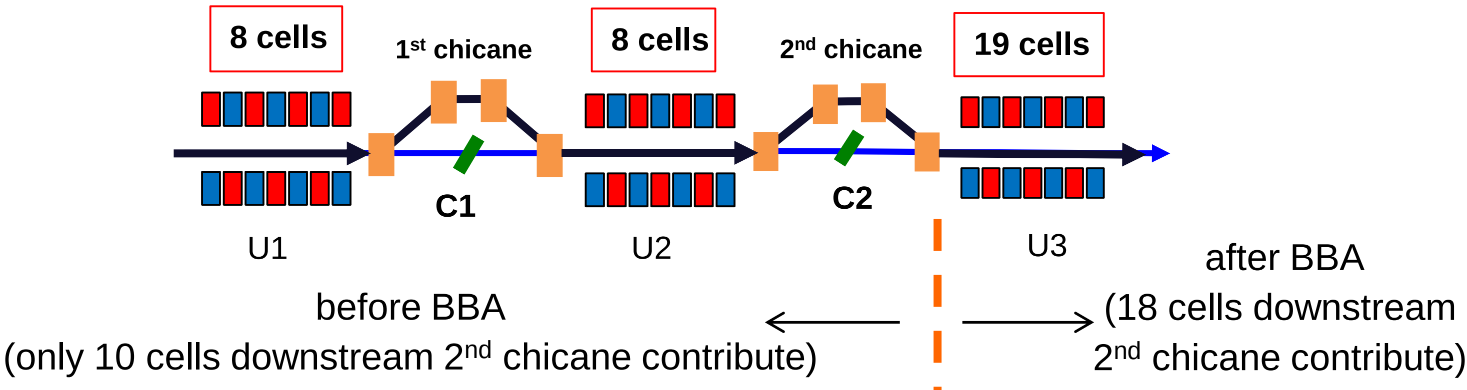
	Photon Energy	Expected min. Pulse Energy
SA1	5-9.3 keV	2 mJ
	>9.3-14 keV	1 mJ
	>14 – 24 keV	0.5 mJ
SA2	5.8-9.3 keV	2 mJ
	>9.3-12 keV	1 mJ
	>12 – 18 keV	0.5 mJ
	>18 – 30 keV	tbd
SA3	0.5-1.5 keV	5 mJ
	>1.5-2.5 keV	2 mJ
	>2.5 keV	tbd

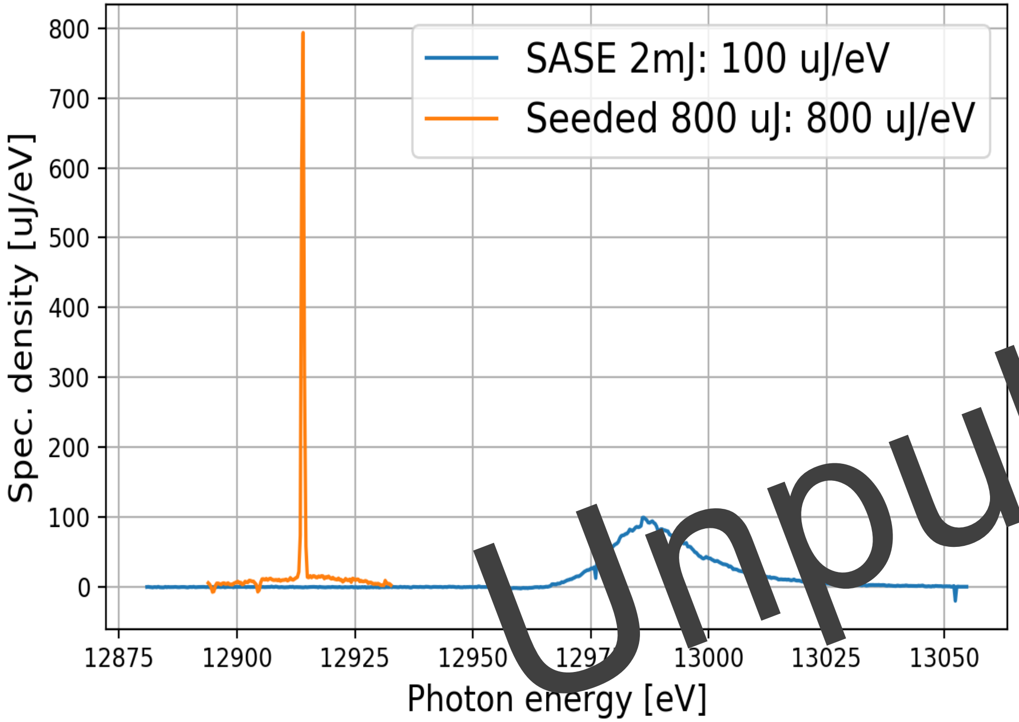


Hard X-Ray Self Seeding

First User Runs

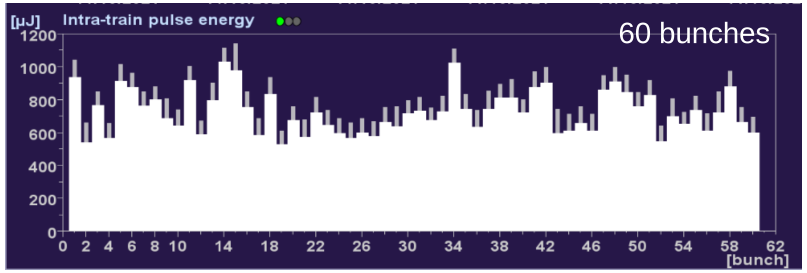
- 3 user runs in Sept. and Oct. with different photon energies
- Set-up time is around 3 shifts (mostly due to challenging trajectory requirements), stable delivery after set-up





	7.5 keV	9 keV	12.9 keV
e- beam energy	11.5 GeV	11.5 GeV	16.5 GeV
SASE performance	1.5 mJ (140 uJ/eV)	1 mJ (40 uJ/eV)	2 mJ (100 uJ/eV)
Seeding with 2nd chicane	-	250 uJ (160 uJ/eV)	300-500 uJ (800 uJ/eV)
Seeding with two chicanes	200 uJ (250 uJ/eV)	-	-
Seeding bandwidth (FWHM)	~0.6 eV	~0.8 eV	~0.6 eV

Multi-bunch with high repetition rate (2.2 MHz)



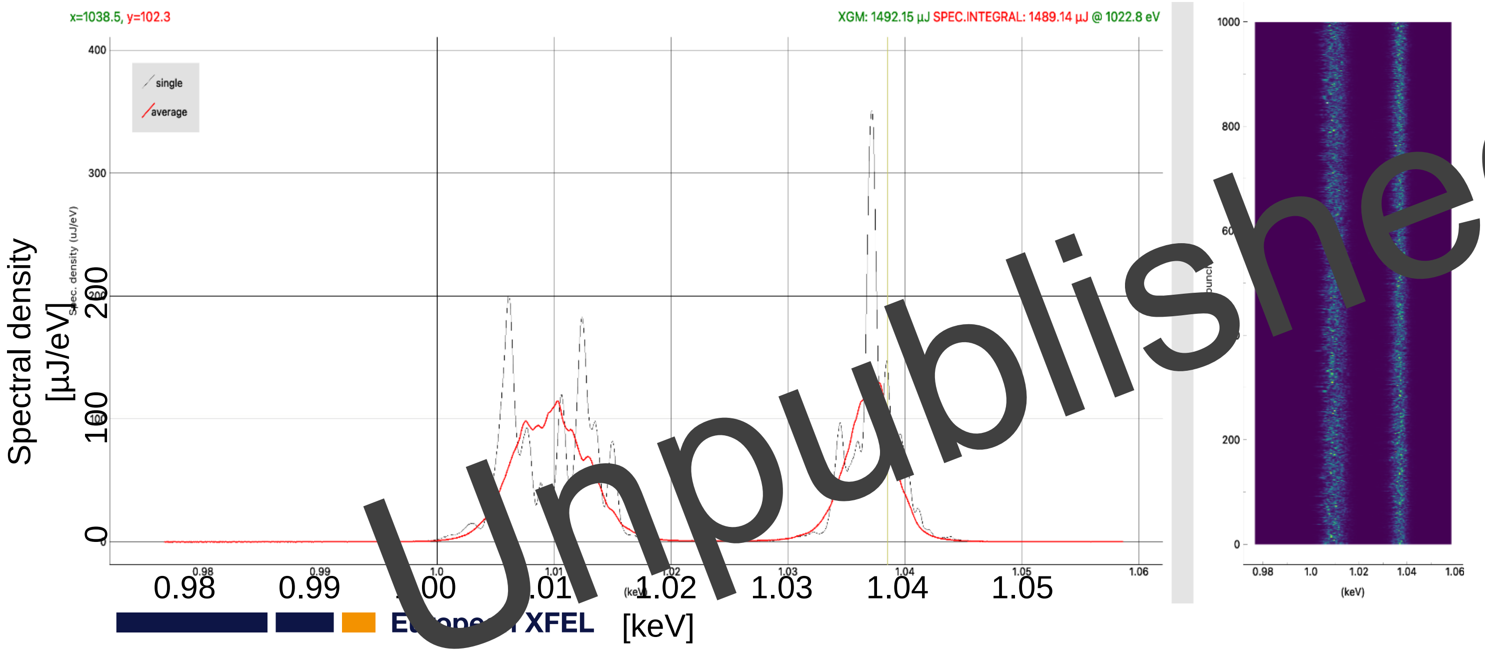
Two colour operation

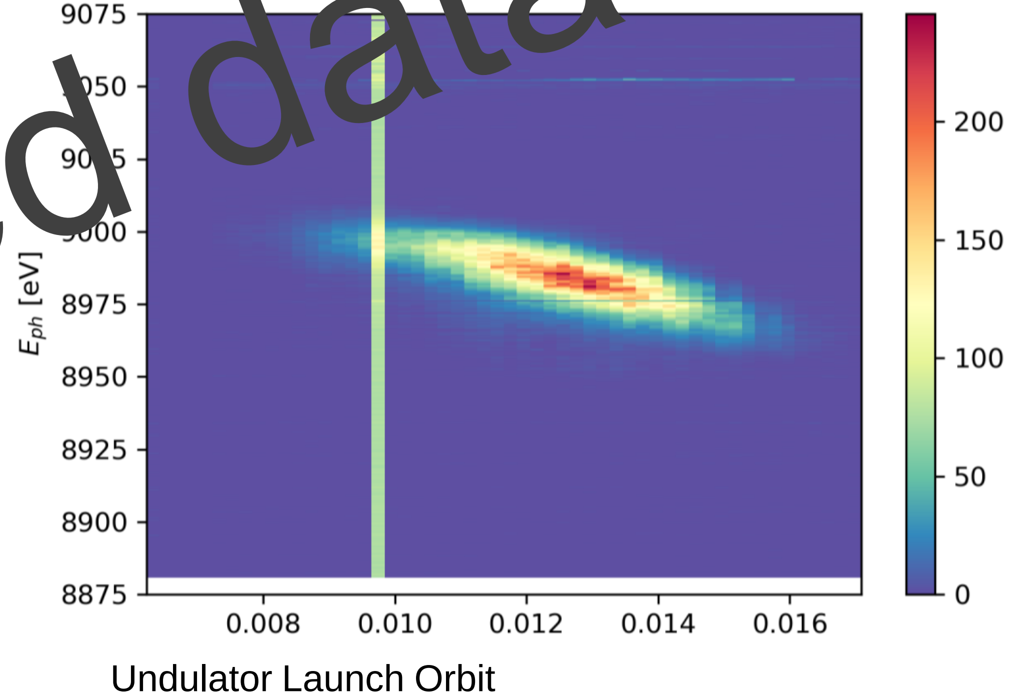
SASE3

- Split undulator setup
  - Freely scannable in color and delay
  - With more than 1mJ/pulse in 700 - 2000 eV
- Dispersion-based fresh-slice for pulse length control
- Regularly delivered on user request to SCS and SQS

SASE2

- Split undulator setup
  - Freely scannable in color and delay
  - With more than 150 uJ/pulse in 7.5 - 10 keV
- Signs of dispersion-based fresh-slice
  - Further work is required

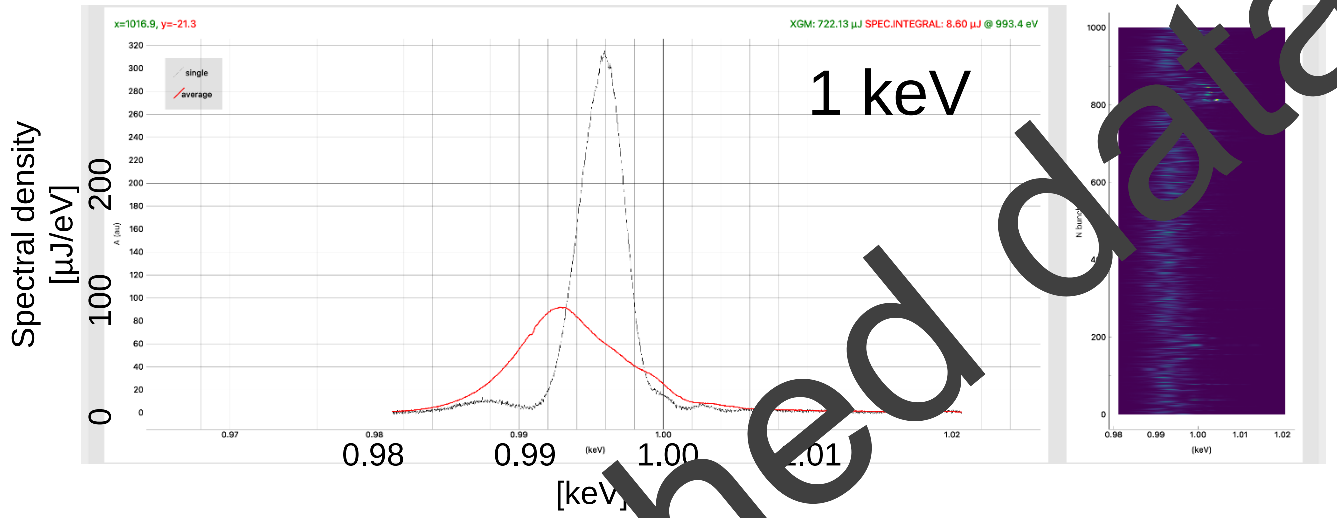


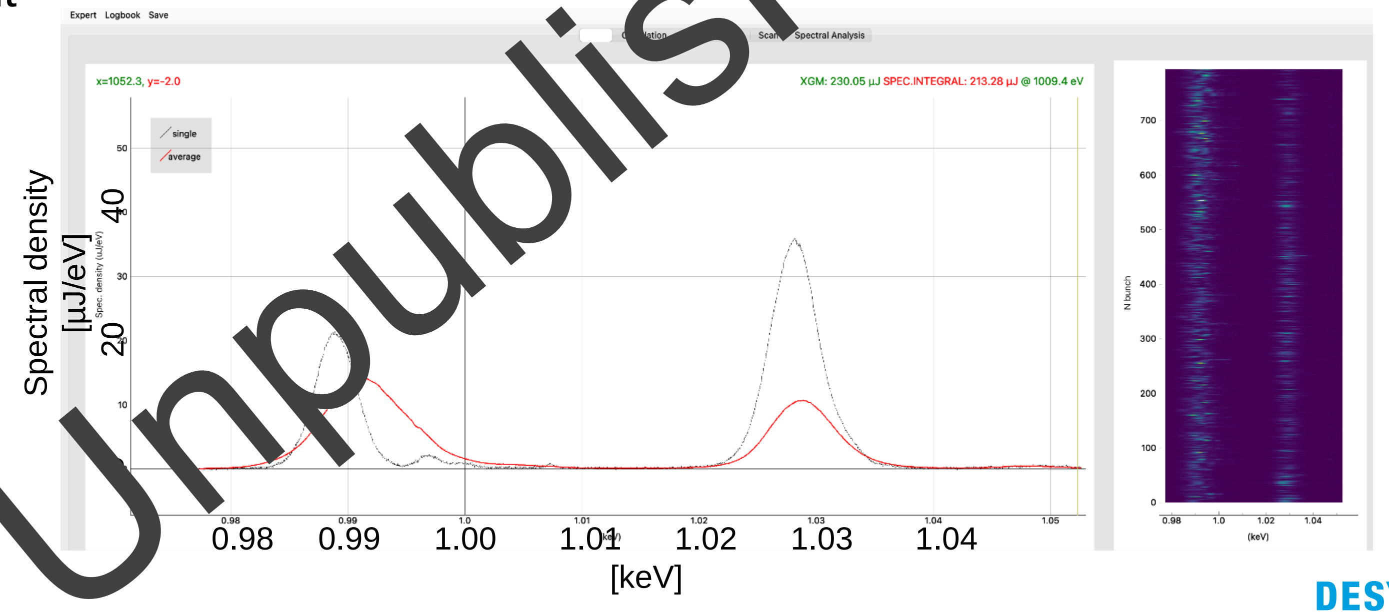


Getting Short

Single Spike Operation at SA3

- Single spectral spikes at SA3
  - Dispersion-Based Fresh-Slice
  - Single spectral spikes
- Production of two single spectral spikes
  - Split undulator setup
  - Two colors are not fully independent





Project History

16 Years Preparation

1993: Start of TESLA technology development

2000: First lasing at 109 nm at the TESLA Test Facility (TTF), now FLASH

2001: TESLA Linear Collider TDR with XFEL appendix

2006: European XFEL TDR

7 Years Construction

2009: Foundation of the European XFEL GmbH

Start of underground construction

2010: Formation of the Accelerator Consortium:

16 accelerator institutes under the coordination of DESY

2012: End of tunnel construction, start of underground installation

2016: Finish of accelerator installation, start of commissioning

5 Years Operation

2017: First lasing SASE1, start of early user program

2018: First lasing SASE3 & SASE2, parallel operation of all 3 FELs

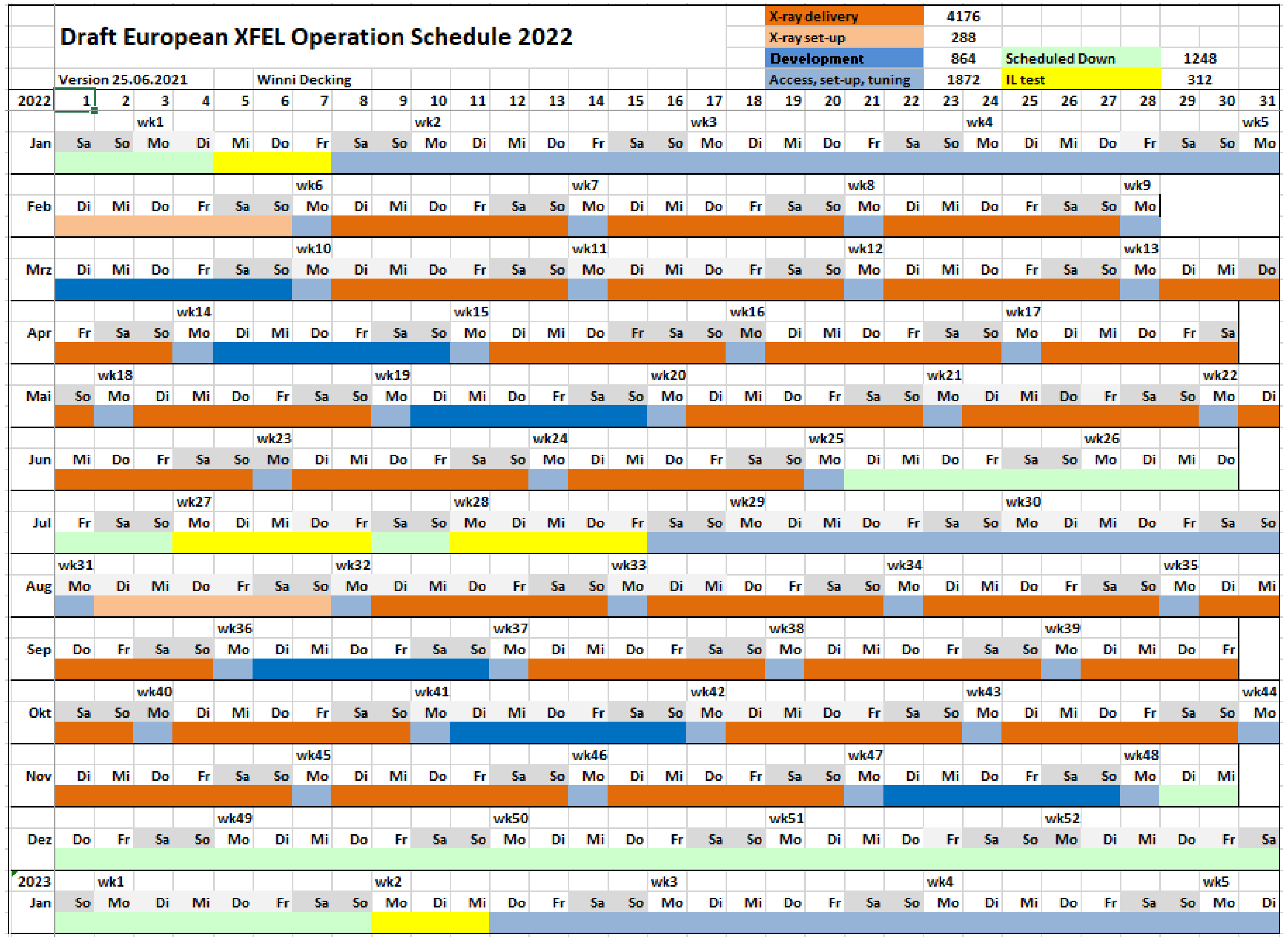
2019: User operation with all 3 FELs and 6 experiments

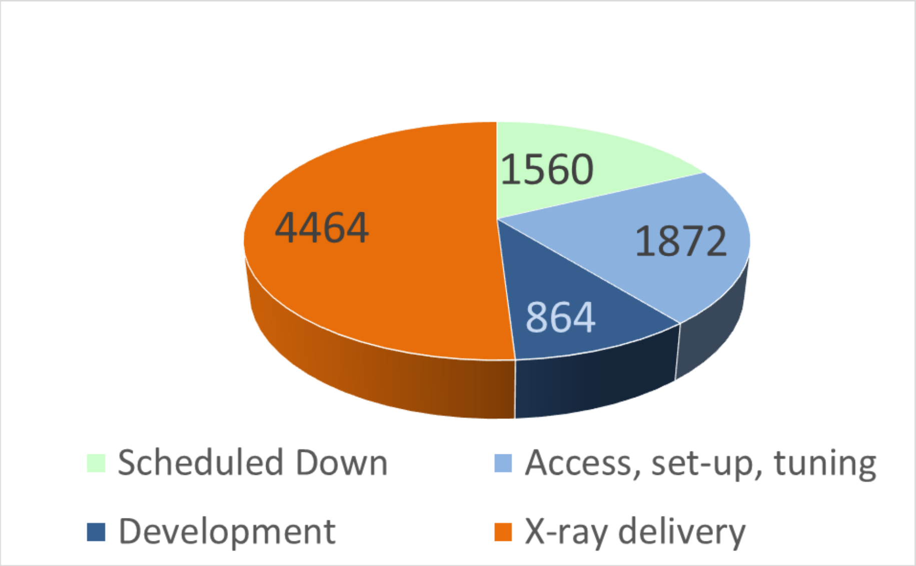
2020: World record photon energy

2021: HXRSS and Two colour operation to users, beam power increases

Outlook

2022 to be a ‘standard’ operation year, but ….





XFEL Accelerator R&D

Towards European XFELs Future

Improve present days operation
add ‘small’ upgrades
and expand operation envelope

Autonomous accelerator
and enhanced user ops

Propose and Execute ‘large’ upgrade
- **More beamlines (2nd fan, SASE4/5)**
- **CW operation**
- **High energy photons**
- **VUV@EuXFEL (?)**

2020

- **AppleX @ SASE3**
- **DemoFELO @ SASE1**
- **Dechirper for diagnostics and short pulses**

2025

- **Gun5**
- **New injector laser**
- **Attosecond capabilities**

- **Superconducting undulators**

2030

2040

Novel accel tech

R&D Program

IMP: Improvement of operational stability and efficiency
CW: Continuous Wave (CW) operation of XFEL
EXT: Extension of the facilities parameters and performance range
IPCROB: Intelligent Process Control & Robotics